Ten-loop fixed-dimension renormalization-group functions of the three-dimensional $O(N)$ model from distributed GPU-accelerated parametric integration

Paolo Ribeca[1, 2]

[1] *Biomathematics and Statistics Scotland (BioSS), Edinburgh, United Kingdom*
[2] *UK Health Security Agency (UKHSA), London, United Kingdom*

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The renormalization-group functions β, η and ν of the three-dimensional $O(N)$ model in fixed dimension, in the massive zero-momentum scheme, were known to eighth order in the coupling. Here we extend them to tenth order, and β to eleventh, for general N—which requires the four-point and two-point functions to ten loops, 1 593 254 graphs at the last order—and we re-evaluate all lower orders independently. Through the Callan–Symanzik equation every coefficient follows algebraically from per-order sums that are linear in finite, zero-momentum $\bar{R}$-subtracted parametric integrals, one per graph; we numerically sample them on GPUs by tropical Monte Carlo, with densities adapted to the forest terms. The computation reproduces the known per-order sums from two to seven loops within three standard deviations, and agrees at ninth and tenth order, at lower precision, with an independent route through the vertex functions of an intermediate scheme. The new coefficients have relative uncertainties of a few 10^{-5} or better, and the eighth-order coefficients of η and ν improve by factors of 120 and 4. An encouraging result concerns the cost: with this strategy, the time needed to integrate an order to a fixed relative precision grows from seven loops on by a factor of only 2.4–2.9 per loop on average. However, the construction of the integrands still grows factorially, and at tenth order it costs more than the integration: 1 200 CPU core-hours against 89 GPU hours. Resummed, the tenth-order series give $\nu = 0.62997(51)$, $\eta = 0.0333(18)$ and $\omega = 0.803(14)$ for $N = 1$, against 0.62997097(12), 0.036297612(48) and 0.82968(23) from the conformal bootstrap; for $N = 2$ and 3 the deviation of ν grows, to -3.1 standard deviations at $N = 3$, where ω lies +10.4 standard deviations from the Monte Carlo value, so that the uncertainty is now dominated by the resummation rather than by the coefficients. Our computational strategy is cooperative and distributed, and thus open to future improvement: every result presented here is generated automatically from, or checked against, a public ledger of independent, exactly pooled runs, hosted on GitHub, to which anyone is welcome to contribute computing time.

I. INTRODUCTION AND STATE OF THE ART

A. Universality and field-theoretic approaches

Continuous phase transitions in systems with short-range interactions and an N-component order parameter with $O(N)$ symmetry fall into the universality classes of the three-dimensional $O(N)$ model: $N = 0$ describes polymers and self-avoiding walks, $N = 1$ uniaxial magnets and the liquid–vapour critical point, $N = 2$ planar magnets and the superfluid transition of ^{4}He, $N = 3$ isotropic magnets [1, 2]. Their critical exponents are a standard benchmark both for field theory and for non-perturbative methods.

Two perturbative field-theoretic approaches have been pushed to high order. In the ε expansion the renormalization-group (RG) functions of ϕ^4 theory are computed by minimal subtraction in $d = 4 - \varepsilon$ and the exponents are expanded in powers of ε [2, 3]. In fixed-dimension perturbation theory, proposed by Parisi [4], one works directly in $d = 3$, where the theory is super-renormalizable, renormalizes it by normalization conditions at zero momentum, and obtains the exponents from the RG functions at the non-trivial zero of the β function. In both approaches the series are asymptotic, with coefficients growing like $k!$ at large order k [5, 6], and must be resummed, typically by a Borel–Leroy transform followed by a conformal mapping built on the known large-order behaviour [7–9]. Fixed dimension has the advantage that Borel summability is established for the correlation functions of the $d = 3$ theory [10], although not directly for the RG functions, while for the ε series it has to be assumed. The precision of the resummed exponents is limited by the number of known terms, by the numerical uncertainty of the highest ones, and by systematic effects of the resummation; in fixed dimension these include the expected non-analyticity of β at the fixed point, caused by confluent corrections to scaling [1, 11].

B. The fixed-dimension series: history and status

We consider the Euclidean $O(N)$ ϕ^4 theory in $d = 3$ in the massive zero-momentum scheme, in which the renormalized mass m, the field normalization and the dimensionless coupling g are fixed by conditions on $\Gamma^{(2)}_R(0)$, $\partial\Gamma^{(2)}_R/\partial p^2(0)$ and $\Gamma^{(4)}_R(0) = m\,g$. We write the series in $\bar{u} = (N + 8)u/6$ with $u = g/8\pi$, i.e., in the normalized coupling of Refs. [12–14] (the $\tilde{u}$ of Ref. [14]). The computation of the coefficient of $\bar{u}^{L+1}$ in β requires the four-point function $\Gamma^{(4)}$ at L loops; the coefficients of $\bar{u}^L$ in η and ν require the two-point function $\Gamma^{(2)}$, with its

p^2 derivative or with ϕ^2 insertions, at L loops.

a. Nickel and collaborators. A numerical evaluation programme was first carried out by Baker, Nickel, Green and Meiron [12], who took the series for $N = 1$ from three to six loops [15], i.e., β to $\bar{u}^7$ and η, ν to $\bar{u}^6$; Baker, Nickel and Meiron [13] analysed the corresponding series for $N = 0$–3. The key technical ingredient was that one-loop bubbles, triangles and boxes in $d = 3$ have closed forms [16], so that each graph could be integrated in momentum space with these one-loop subgraphs replaced by effective vertices. The individual graph values were compiled, together with their weights, in an unpublished Guelph report [17]. The 1042 diagrams needed through six loops [15] are given to ten decimals when the integral has dimension at most three and to eight decimals when it has dimension five or six, with errors "generally less than ± 5 in the last digit quoted" and up to ± 20 in exceptional cases. The whole evaluation took about 20 hours on an IBM 370/155 and 3 hours on a CDC 6600 [17]. The integral of each graph was set up by hand, and the paper on the numerical methods announced in Refs. [12, 13, 17] never appeared. Because the compilation separates integrals from symmetry factors, it was reused for decades to build the series of other models without recomputing any integral, for instance for cubic anisotropy [18]; see Ref. [1] for a review. Murray and Nickel later extended η and η_{ϕ^2}, hence ν, to seven loops for $N = 0$–3 [19]; this report is also unpublished, its method is undocumented, and its coefficients are quoted, to about seven significant digits, in the appendix of Ref. [9]. The resummation of these series by Le Guillou and Zinn-Justin and by Guida and Zinn-Justin [7–9] produced the standard fixed-dimension estimates of the $O(N)$ exponents.

b. Automation. By 2003 the six-loop integrals had not been reproduced by an independent group for a quarter of a century, and the next order involves several thousand graphs. References [15, 20] therefore set out to automate the whole computation: graph generation, weights and $O(N)$ factors, and an automatic search for the cheapest effective-vertex parametrization of every amplitude, followed by deterministic adaptive cubature. All seven-loop amplitudes were parametrized with at most nine integration variables [15], but the seven-loop numerical evaluation was not completed at the time.

c. Phi4tools. Sberveglieri and Spada [14] (the Phi4tools compilation) recomputed all zero-momentum graphs with up to eight vertices, that is, $\Gamma^{(4)}$ to seven loops and $\Gamma^{(2)}$ to eight. They essentially followed the strategy proposed in Ref. [20], integrating every graph in momentum space after automatically assigning effective vertices, with only slight differences in their own words: separate tools for each step, Mathematica for the integrals of dimension up to three and the VEGAS algorithm [21] for the rest in place of deterministic adaptive cubature, some new analytic and numerical effective vertices, and a simpler search for the parametrization of the amplitudes. Some of the hardest graphs were cross-checked with feyntrop [22], an implementation of tropical Monte Carlo. Their largest relative uncertainties are a few parts in 10^5 for two- and four-point graphs and a few parts in 10^4 for the p^2 derivatives of two-point graphs. This provided the first independent check of Ref. [17]: of the 585 six-loop four-point graphs renormalized in the same way, most values agree to ± 1 in the last digit quoted by Nickel et al., with few exceptions, of which only one differs by about 20 units. From these values they obtained β, η and ν to $\bar{u}^8$ for general N, and the corresponding series of the cubic model. For $N = 1$ the quoted uncertainties of the $\bar{u}^8$ coefficients, added linearly over the coefficients of the N polynomials, are about 4×10^{-6} (β), 5×10^{-4} (η) and 5×10^{-5} (ν) in relative terms. The authors postponed the resummation to future work [14]; to our knowledge the eighth-order fixed-dimension series have not been resummed. Table I summarizes this history.

C. The ε expansion and non-perturbative benchmarks

In the ε expansion the minimal-subtraction RG functions of the $O(N)$ model were extended to five loops [24–27] and to six [28, 29], the latter computation also estimating the graphs without subdivergences to eleven loops and comparing them with the large-order behaviour of β. Position-space methods based on graphical functions have since given the β function and the anomalous dimensions to seven loops [23], for the $O(N)$ model as well (published for $N = 1$, the $O(N)$ results distributed with the author's software), and for $N = 1$ the field anomalous dimension to eight [30]. Numerical approaches in minimal subtraction remove subdivergences by subtraction operators acting on the integrands, so that the RG functions follow algebraically from linear sums of finite integrals [31, 32]; the zero-momentum $\bar{R}$ operation used here (Sec. II A 3) is the fixed-dimension counterpart of this idea. The six-loop ε series have been resummed for general N [29, 33], and for $N = 1$ the seven-loop series and, for η, the eight-loop one [34]. A recent study for $N = 1$ [34] finds that the uncertainty of η decreases rapidly with the loop order while those of ν and of the correction-to-scaling exponent ω do not, and reports a slight but systematic tension with the conformal bootstrap.

The most precise determinations of the exponents are now non-perturbative. The numerical conformal bootstrap determines the Ising exponents with uncertainties far below those of any resummed perturbative series, and constrains the $O(N)$ exponents as well [35, 36]; Monte Carlo simulations, of the Ising model [37] and of improved lattice Hamiltonians [38–40], also reach high precision. The conformal bootstrap gives $\nu = 0.62997097(12)$ and $\eta = 0.036297612(48)$ for $N = 1$ [41], $\nu = 0.67175(10)$ and $\eta = 0.038176(44)$ for $N = 2$ [42], and $\nu = 0.71168(41)$ and $\eta = 0.03787(13)$ for $N = 3$ [43], in agreement with the Monte Carlo values $\nu = 0.67169(7)$, $\eta = 0.03810(8)$

TABLE I. Perturbative RG functions of the $O(N)$ model: highest known orders. In fixed dimension the term $\bar{u}^k$ of β needs $\Gamma^{(4)}$ at $L = k - 1$ loops, and that of η and ν needs $\Gamma^{(2)}$ at $L = k$ loops; the graph counts are one-particle-irreducible (1PI) graphs without self-loops, one-vertex-reducible ones (with a cut vertex) included. The precision is the relative uncertainty of the highest coefficient at $N = 1$ or, where no uncertainty is quoted, the number of significant digits printed (sig.) or the absolute accuracy of the individual integrals (int.). The ε-expansion coefficients are known exactly, as rational combinations of multiple zeta values and, at seven loops, of one further period that is conjecturally not of this form [23].

Function	Term	Graphs	Reference	Precision
Fixed dimension $d = 3$, massive zero-momentum scheme				
β	$\bar{u}^7$	$\Gamma^{(4)}$, $L = 6$ (660)	[12] ($N = 1$); [13, 17] ($N = 0$–3)	6 sig.; int. 10^{-8}–10^{-10}
	$\bar{u}^8$	$\Gamma^{(4)}$, $L = 7$ (3986)	[14], all N	4×10^{-6}
	$\bar{u}^9$	$\Gamma^{(4)}$, $L = 8$ (26 540)	this work, all N	2×10^{-6}
	$\bar{u}^{10}$	$\Gamma^{(4)}$, $L = 9$ (197 022)	this work, all N	1×10^{-5}
	$\bar{u}^{11}$	$\Gamma^{(4)}$, $L = 10$ (1 593 254)	this work, all N	1×10^{-5}
η, ν	$\bar{u}^6$	$\Gamma^{(2)}$, $L = 6$ (75)	[12] ($N = 1$); [13, 17] ($N = 0$–3)	int. 10^{-8}–10^{-10}
	$\bar{u}^7$	$\Gamma^{(2)}$, $L = 7$ (317)	[9, 19], $N = 0$–3	~ 7 sig.
	$\bar{u}^8$	$\Gamma^{(2)}$, $L = 8$ (1622)	[14], all N	5×10^{-4} (η), 5×10^{-5} (ν)
	$\bar{u}^9$	$\Gamma^{(2)}$, $L = 9$ (9184)	this work, all N	1×10^{-5} (η), 1×10^{-5} (ν)
	$\bar{u}^{10}$	$\Gamma^{(2)}$, $L = 10$ (59 614)	this work, all N	1×10^{-5} (η), 4×10^{-5} (ν)
ε expansion, minimal subtraction: β, γ_{m^2}, γ_ϕ				
	5 loops		[3, 24–27], all N	exact
	6 loops		[28, 29], all N	exact
	7 loops		[23, 30], all N	exact
	8 loops	γ_ϕ only	[30], $N = 1$	exact

for $N = 2$ [39] and $\nu = 0.71164(10)$, $\eta = 0.03784(5)$ for $N = 3$ [40]. For $N = 1$ in particular, these results make the perturbative series a stringent test of resummation methods, a test that is only meaningful if the series coefficients and their uncertainties are reliable.

D. This work

Since 1978 each fixed-dimension series has been extended by one order at a time [14, 19]. Both complete computations [14, 17] used one momentum-space representation per graph, chosen by hand or automatically and integrated to high accuracy, and obtained the RG functions from the vertex functions of an intermediate scheme through the Z factors (Nickel et al. subtracted, according to Ref. [14], all two-point subgraphs at zero momentum, Phi4tools only the divergent tadpole and sunset subgraphs). The ninth and tenth orders need 26 540 and 197 022 four-point graphs at eight and nine loops, and 9184 and 59 614 two-point graphs at nine and ten loops, against 3986 and 1622 at the eighth order, each of higher dimension; the eleventh order of β needs 1 593 254 four-point graphs at ten loops. Moreover, along this route the order-L contributions largely cancel: at four, five and six loops the intermediate-scheme four-point input to β exceeds the net order-L four-point coefficient B_L (Sec. II) by factors of about 29, 50 and 74, so that its relative error must be smaller than the target by the same factor.

In this paper we compute β, η and ν to $\bar{u}^{10}$ (β to $\bar{u}^{11}$) for general N, and re-evaluate all lower orders, with a method designed for this scale and for massively parallel hardware: its precision depends as little as possible on the order, and its cost should grow slowly with the order (Sec. II C 1); measured, from seven loops on, the cost grows by a factor of 2.4–2.9 per loop (Sec. III B).

1. Through the Callan–Symanzik equation, the coefficients of β, η and ν follow algebraically from three functions, $\Gamma_R^{(4,1)}(0)$, $\partial\Gamma_R^{(2,1)}/\partial p^2(0)$ and $\Gamma_R^{(2,2)}(0)$, whose L-loop terms we call B_L, H_L and C_L (Sec. II A 1). Each is a *linear* sum over graphs of finite integrals, subtracted at zero momentum by Zimmermann's forest formula [44–47], the $\bar{R}$ operation of Sec. II A 3. No Z factors and no series reversion are needed, and the cancellations that the Z-factor route performs between graphs take place inside each integrand. We call this route A.

2. At zero external momenta and unit masses every graph, with all its forest terms, is a single parametric integral over a simplex, built from the Kirchhoff polynomials of the graph and of its subgraphs and quotient graphs [48–50]. Every integrand is sampled by tropical, Hepp-sector Monte Carlo [22, 51–54], with a density adapted to the forest terms, on any number of CPUs or GPUs from the same source code.

3. The computation is automatic, from graph generation to series coefficients: there is no per-graph

choice of momentum routing, effective vertices or integration variables. Since every graph carries its $O(N)$ polynomial, one run gives every N.

4. The results are checked against exact low-order values, against the published series, against the large-N limit and the large-order behaviour, and, at the ninth and tenth orders, against a second, independent route (route B) through the vertex functions of an intermediate scheme, its Z factors and the reversion of the coupling series. The precision of route B, 7×10^{-4}–7×10^{-3} relative, is far coarser than that of the new coefficients, and the coefficient of $\bar{u}^{11}$ in β is checked only by the internal statistics, the large-N limit and the large-order behaviour.

The results are therefore an independent determination of all known orders: the renormalization route, the integral representation and the numerics share nothing with Refs. [14, 17] except the graph lists, which our own generator reproduces wherever both exist (Sec. II B 3). The published coefficients combine our new sums with, at each lower order, the more precise of our sum and the one derived from Ref. [14]; with our own lower orders only, they change by at most 0.0075 standard deviations (Sec. III C).

Runs on NVIDIA H200 GPUs reproduce the known per-order sums from two to seven loops for all three functions within three standard deviations, with error estimates that the within-run statistics confirm to a few per cent (Sec. III A). At six loops, with the computing time we allocated to it, the method reproduces the published values with similar or better precision. From seven loops on, where the graphs are many and their integrals of high dimension, it overtakes the per-graph treatment: all three seven-loop sums are now known more precisely than from Phi4tools, by factors of about 1.5, 11 and 2.9 for B_7, H_7 and C_7, and the complete sum H_7, which fixes the $\bar{u}^7$ term of η (209 integrands), reaches a relative uncertainty of 10^{-5} in about 6.6 GPU hours, against 5.5×10^{-5} for the same quantity derived from the Phi4tools series (Sec. III). The new coefficients, from ninth to eleventh order, have relative uncertainties between 2×10^{-6} and 4×10^{-5} at $N=1$ (Tables XI–XIII), and the coefficient of $\bar{u}^8$ in η is now 120 times more precise than before (Sec. III C). The uncertainties thus hardly grow with the order. Along the per-graph route the p^2 derivatives are the least precise: at the eighth order of Ref. [14], 5×10^{-4} for η against 4×10^{-6} for β (Table I). A uniform precision ensures that the number of known orders, not their statistical errors, limits the resummation: at the present orders the statistical errors are already negligible there (Sec. II D 3), while the precision of the highest coefficients matters for the large-order and consistency tests and for the orders still to come.

The paper is organized as follows. Section II describes the series, the graphs, the numerical method and the resummation; Sec. III gives the validation, the performance, the new coefficients and the exponents, compared with other determinations; Sec. IV summarizes. Five appendices give the conventions, the graphs, the zero-dimensional sum rules, the reproducibility checks, and the availability of the data and the code.

II. METHODS

This section explains how the series coefficients are reduced to integrals (Sec. II A), how the graphs and their integrands are built (Sec. II B), how the integrals are evaluated (Sec. II C), how the series are resummed (Sec. II D), and how the results are checked (Sec. II E). A claim that a script of the repository verifies is marked with the identifier of its check, as in [check G2]; Appendix D lists the checks.

A. Computing the series

1. *Scheme and Callan–Symanzik relations*

We consider the Euclidean action

$$S[\phi]=\int d^3x\,\Big[\tfrac{1}{2}(\partial_\mu\phi)^2+\tfrac{1}{2}m_0^2\,\phi^2+\tfrac{\lambda_0}{4!}\,(\phi^2)^2\Big] \qquad (1)$$

for an N-component field ϕ, and denote by $\Gamma^{(a,b)}$ the one-particle-irreducible (1PI) vertex functions with a legs and b insertions of $\frac{1}{2}\phi^2$ at zero momentum, with the $O(N)$ tensors factored out [2, 15]. In the massive zero-momentum scheme [4, 12, 13] the renormalized mass m, the normalizations of ϕ and ϕ^2, and the dimensionless coupling g are fixed by

$$\Gamma_R^{(2)}(p)=m^2+p^2+O(p^4),$$
$$\Gamma_R^{(4)}(0)=m\,g,\qquad \Gamma_R^{(2,1)}(0;0)=1. \qquad (2)$$

The renormalized functions obey the Callan–Symanzik equation [2, 15]

$$\Big[m\,\partial_m+\beta\,\partial_g-\tfrac{a}{2}\,\eta-b\,(\eta_{\phi^2}-\eta)\Big]\Gamma_R^{(a,b)}=(2-\eta)\,m^2\,\Gamma_R^{(a,b+1)} \qquad (3)$$

for $b\le1$, where the right-hand side carries one more insertion at zero momentum, and $\beta=m\,\partial g/\partial m$, $\eta=m\,\partial\ln Z/\partial m$ and $\eta_{\phi^2}=m\,\partial\ln Z_{\phi^2}/\partial m$ are taken at fixed bare parameters (Z and Z_{ϕ^2} normalize ϕ and ϕ^2). Applied at zero momenta to the conditions (2), the first through its p^2 derivative, Eq. (3) gives, with $u=g/8\pi$ and $\beta_u=m\,\partial u/\partial m$,

$$\eta=\frac{2h}{1+h},\qquad \beta_u=-u+2\eta\,u+(2-\eta)\,\tilde{b},$$
$$\eta_{\phi^2}=-(2-\eta)\,c,\qquad \frac{1}{\nu}=(2-\eta)(1-c), \qquad (4)$$

where

$$\tilde{b}=\frac{m\,\Gamma_R^{(4,1)}(0)}{8\pi},\qquad c=m^2\,\Gamma_R^{(2,2)}(0),$$

$$h = -m^2 \left.\frac{\partial \Gamma_R^{(2,1)}}{\partial p^2}\right|_{p=0}. \tag{5}$$

The exponents follow at the non-trivial zero of β_u (Sec. II D).

The three functions (5) are renormalized vertex functions with insertions, superficially convergent, and their expansions in u,

$$\tilde{b} = \sum_{L\geq 1} (-1)^{L+1} B_L\, u^{L+1},$$
$$h = \sum_{L\geq 2} (-1)^{L} H_L\, u^{L}, \qquad c = \sum_{L\geq 1} (-1)^{L+1} C_L\, u^{L}, \tag{6}$$

are *linear* in the graphs: B_L is a weighted sum over the L-loop graphs of $\Gamma^{(4)}$ with one insertion, H_L and C_L over the L-loop graphs of $\Gamma^{(2)}$ with one and two insertions, of integrals in which every renormalization part is subtracted at zero momentum (Secs. II A 2 and II A 3). No Z factor appears, because the subtractions implement the conditions (2) graph by graph. Together with lower orders, the coefficient of $\bar{u}^{L+1}$ in β is fixed by B_L and, through the term $2\eta u$, by H_L; that of $\bar{u}^L$ in η by H_L; and that of $\bar{u}^L$ in ν by C_L and H_L [Eq. (A2) in Appendix A]. Only products and quotients of series enter Eq. (4); no series has to be reversed. At lowest order, for instance, $B_1 = (N+8)/12$ gives $\beta(\bar{u}) = -\bar{u} + \bar{u}^2 + O(\bar{u}^3)$ for every N. We write the RG functions in the normalization of Ref. [14],

$$\beta(\bar{u}) = -\bar{u} + \bar{u}^2 + \sum_{k\geq 3} \frac{p_{\beta,k}(N)}{(N+8)^{k-1}}\, \bar{u}^k,$$
$$\eta(\bar{u}) = \sum_{k\geq 2} \frac{p_{\eta,k}(N)}{(N+8)^{k}}\, \bar{u}^k, \tag{7}$$
$$\eta_{\phi^2}(\bar{u}) = \sum_{k\geq 1} \frac{p_{\eta_{\phi^2},k}(N)}{(N+8)^{k}}\, \bar{u}^k,$$
$$\nu(\bar{u}) = \frac{1}{2} + \sum_{k\geq 1} \frac{p_{\nu,k}(N)}{(N+8)^{k}}\, \bar{u}^k,$$

with polynomials $p_{X,k}(N)$, which follow from the polynomials $B_L(N)$, $H_L(N)$, $C_L(N)$ through Eq. (4) applied to the polynomials themselves, in exact rational arithmetic, rather than to their values at integer N, whose interpolation is ill-conditioned. Equation (4) is the fixed-dimension counterpart of the representation of the minimal-subtraction RG functions by finite subtracted integrals [31, 32]. We call this computation, through Eqs. (4)–(6), route A.

The previous complete computations [13, 14, 17] and the automated approach of Ref. [15] instead evaluated the zero-momentum vertex functions of a scheme in which only two-point subgraphs are subtracted (in Refs. [14, 15] only the tadpole, set to zero, and the sunset, subtracted at $p^2 = 0$; in Ref. [17], according to Ref. [14], all two-point subgraphs), and obtained the RG functions through the Z factors, the relation between the coupling of that scheme and the renormalized one, and its inverse [15]; we use this Z-factor route as an independent check (route B, Sec. II E). Along that route, in the intermediate scheme of Refs. [14, 15], the contribution of the L-loop graphs is a small difference of large numbers: the L-loop four-point graphs contribute to the coefficient of β an amount larger than the net B_L by factors of 5.5, 14.4, 29.2, 49.5 and 74 at $L = 2, \ldots, 6$, and the L-loop graphs of $\partial\Gamma^{(2)}/\partial p^2$ contribute to η an amount larger than H_L by about 20, 114 and 106 at $L = 4$, 5 and 6; the rest is cancelled by products of lower orders. The relative error of the coefficient is the relative error of the graph sum multiplied by this factor, so that the number of samples grows with its square. In Eq. (6) the same cancellations take place within each integrand, between a graph and its counterterms evaluated at the same point, so that they cost no additional samples; Sec. III B quantifies the gain.

2. *Parametric integrands*

At zero external momenta and unit mass, the integral of a graph G with ℓ lines and L loops, with one d^3k/π^2 per loop (Nickel's normalization [17]), is

$$A_G = \frac{\Gamma(\ell - 3L/2)}{\pi^{L/2}} \int_\Delta d\mu\, U_G^{-3/2}, \quad U_G = \sum_T \prod_{e\notin T} x_e, \tag{8}$$

where T runs over the spanning trees of G, Δ is the simplex $x_e \geq 0$, $\sum_e x_e = 1$, and $d\mu = \delta(1 - \sum_e x_e) \prod_e dx_e$ [15, 48, 50]; the one-loop bubble, for instance, equals one. The Kirchhoff polynomial U_G factorizes over the 2-connected components (blocks) of G. An insertion of $\frac{1}{2}\phi^2$ at zero momentum on line e squares its propagator, that is, it multiplies the Schwinger integrand by the parameter of e: in Eq. (8) it adds a factor x_e and raises the argument of the Gamma function by one. For a two-point graph with external vertices a and b, the external momentum enters as $\exp(-p^2\, W_G/U_G)$, where W_G is the Kirchhoff polynomial of the graph with a and b identified (equivalently, the polynomial of the spanning 2-forests that separate a from b; $W_G = 0$ if $a = b$). The p^2 derivative at $p = 0$ is therefore $-\Gamma(\ell - 3L/2 + 1)\, \pi^{-L/2} \int_\Delta d\mu\, W_G\, U_G^{-5/2}$, a single one-signed integrand, which replaces the expansion into $n(n-1)/2 + 2n$ signed terms per chain of n propagators used in momentum space [15].

Since $\ell = 2L$ for $\Gamma^{(4)}$ and $\ell = 2L - 1$ for $\Gamma^{(2)}$, all integrals entering Eq. (6) carry the same prefactor $\Pi_L = \Gamma(L/2 + 1)\, \pi^{-L/2}$, and the per-order sums are

$$B_L = \Pi_L \sum_{G\in\mathcal{G}_4(L)} S_G \sum_e \int_\Delta d\mu\, x_e\, \bar{R}_{G,e}\big[U_G^{-3/2}\big],$$
$$H_L = \Pi_L \sum_{G\in\mathcal{G}_2(L)} S_G \sum_e \int_\Delta d\mu\, x_e\, \bar{R}_{G,e}\big[W_G\, U_G^{-5/2}\big], \tag{9}$$

$$C_L = \Pi_L \sum_{G\in\mathcal{G}_2(L)} S_G \sum_{(e,e')} \int_\Delta d\mu\; x_e\, x_{e'}\, \bar{R}_{G,ee'}\big[U_G^{-3/2}\big],$$

where $\mathcal{G}_E(L)$ is the list of L-loop graphs with E legs, S_G the weight of G including its $O(N)$ factor (Sec. II B), e runs over the lines and (e,e') over the ordered pairs of lines, $e=e'$ included, and $\bar{R}_{G,e}$ is the subtraction operator of Sec. II A 3, which depends on where the insertions are. The propagators have the renormalized mass, $m=1$. (Refs. [14, 15, 17] use propagators of unit mass in the scheme of their integrals instead; individual integrals differ, the RG functions do not.) The sums are known exactly at one and two loops [Eq. (A3)]. On the simplex the integrands can be multiplied by $(\sum_e x_e)^{-(L/2+1)}$, which equals one there and makes them homogeneous of degree $-\ell$: the integrals are then projective, and can be evaluated in any coordinates of the projective simplex (Sec. II C 2). The integration dimension is $\ell-1$, that is 15 for the eight-loop four-point and 16 for the nine-loop two-point graphs.

3. *Zero-momentum subtractions*

The renormalization parts of a graph are its 1PI subgraphs γ of three types, classified by their legs (the lines and external legs of G attached to γ) and by the insertions on their lines: four-point parts without insertions, subtracted by t^0_γ, the term of order zero of the Taylor expansion in the external momenta of γ; two-point parts without insertions, subtracted by t^2_γ, the terms up to order two (mass and field normalization); and two-point parts with one insertion, subtracted by t^0_γ. These subtractions enforce the conditions (2) graph by graph [44–47, 49]. In $d=3$ only the tadpole and the sunset subgraphs need renormalization [14, 15]; all the other subtractions are finite, but the scheme requires them, and through them the cancellations described above take place inside each integrand. The graphs of Eq. (9) are themselves not renormalization parts, except those of H_L, whose own subtraction t^0 does not depend on p^2 and drops out of the derivative, so that the subtraction of the subgraphs alone, $\bar{R}$, is the complete renormalization:

$$\bar{R}_G = \sum_F \prod_{\gamma\in F} (-t_\gamma), \tag{10}$$

where F runs over the forests, sets of renormalization parts that are pairwise nested or vertex-disjoint (the compatibility condition of the counterterms generated by a Lagrangian), the empty forest included, and inner operators act first.

In parametric space the subtractions have closed forms. The term t^0_γ replaces U_θ, in the parent θ of γ, by $U_\gamma\, U_{\theta/\gamma}$, where θ/γ is θ with γ contracted to a point: this is the leading term of U_θ when the parameters of γ go to zero. For a two-point part γ joined to the rest at the vertices a and b, the classical factorization of the Kirchhoff polynomial across a two-vertex cut [55, 56],

$$\begin{aligned} U_\theta &= U_\gamma\, U_{\theta/\gamma} + W_\gamma\, U_{\theta\setminus\gamma} \\ &= U_\gamma\,\big(U_{\theta/\gamma} + \varepsilon_\gamma\, U_{\theta\setminus\gamma}\big), \qquad \varepsilon_\gamma = \frac{W_\gamma}{U_\gamma}, \end{aligned} \tag{11}$$

where $\theta\setminus\gamma$ is θ without the lines of γ and W_γ is the Kirchhoff polynomial of γ with a and b identified (a spanning tree of θ contains either a spanning tree of γ, and then a spanning tree of θ/γ, or a spanning two-forest of γ that separates a from b, and then its other lines connect a and b through $\theta\setminus\gamma$), shows that γ acts on its parent as a massless line with Schwinger parameter ε_γ; t^2_γ keeps the terms of order zero and one in ε_γ, and reduces to t^0_γ when $a=b$. The term of a forest F is thus

$$T_F = (-1)^{|F|}\, x_{\rm ins} \prod_{\theta\in F\cup\{G\}} U_{\theta/F}^{-3/2}, \tag{12}$$

where $x_{\rm ins}$ is the monomial of the insertions and θ/F is the minor of θ in which every maximal element of F below θ is contracted (t^0) or replaced by its effective line (t^2), with the result expanded to first order in every ε; for H_L the factor of the root is $W_{G/F}\, U_{G/F}^{-5/2}$. Both terms of Eq. (11) have the degree of U_θ, so all forest terms are homogeneous of the same degree: they share the simplex and the prefactor of Eq. (9), and their sum, with all positions of the insertions, is integrated as one function [50].

B. Graphs and integrands

The pipeline from the Lagrangian to the sampled integrands has three stages, each with its own checks: for each order, the generation of the graphs, the construction of the subtracted integrand of every graph, and the consistency checks that tie the two to exact results. All three are automatic and were run here to ten loops.

1. *Generating the graphs*

The lists of 1PI graphs of $\Gamma^{(2)}$ and $\Gamma^{(4)}$ are produced by a generator that derives from the one written for the seven-loop computation of Ref. [15], with three improvements: it grows the graphs by vertex insertion instead of enumerating labelled graphs and screening them by a canonical form, it canonicalizes them by colour refinement with backtracking, and it takes any list of vertex degrees, the external legs being implicit in the valences. The class it generates is every connected graph with the given valences, self-loops and bridges allowed, and the construction is complete: removing a leaf of a spanning tree and re-pairing the half-lines it leaves is the inverse of an insertion, so that every graph with V vertices descends from one with $V-1$. The class is grown order by order, each candidate canonicalized and kept in a hash set, and

TABLE II. Generation of the graph lists: numbers of 1PI graphs without self-loops (one-vertex-reducible ones included), and the CPU time of the generator on one core of a cluster node (an Intel Xeon Platinum 9242), one list at a time on an otherwise idle node, the Nickel indices, weights and sum rules included. The eleven- and twelve-loop lists were made on 96 threads, in 23 s and 275 s of wall time for $\Gamma^{(2)}$ and 1005 s for $\Gamma^{(4)}$ at eleven loops, the last with 38 GB of memory; their times in the table are core times summed over the threads. Every regenerated list equals the stored one. The counts of graphs whose integrands survive the pruning (Sec. II B 2) are in Table XVI.

L	$\Gamma^{(4)}$	time	$\Gamma^{(2)}$	time
7	3986	2.1 s	317	0.1 s
8	26 540	22 s	1622	1.1 s
9	197 022	4.3 min	9184	9.8 s
10	1 593 254	53 min	59 614	98 s
11	13 975 511	14.1 h	428 164	22 min
12	—	—	3 393 754	4.2 h

the 1PI graphs are written out at the end of every order, together with their Nickel indices and weights. The cost is a few hundred insertions per graph grown, so that it follows the size of the class, growing by a factor of ten to twelve per loop on one core. Table II gives the sizes and the times, measured on one core of a cluster node; the counts through eight loops are those of Ref. [15] (Table 3.1), the others are new [check G1]. Every complete list satisfies its zero-dimensional sum rule (Sec. II B 3) through twelve loops for $\Gamma^{(2)}$ and eleven for $\Gamma^{(4)}$.

Every graph is converted to its Nickel index [17] in canonical form—the lexicographic minimum over vertex labellings—which identifies it across lists and keys its random numbers (Sec. II C 5); one-vertex-reducible graphs and graphs with both legs on one vertex are flagged, since their integrands vanish (Sec. II B 2), and are not built for B_L and H_L.

a. Weights. Each graph carries the weight $S_G = T_G(N)\,E!/|\mathrm{Aut}\,G|$, where E is its number of legs, $\mathrm{Aut}\,G$ its automorphism group at the level of half-edges with unlabelled legs, and $T_G(N)$ its $O(N)$ factor, obtained by contracting the vertex tensors $(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})/3$ over the 3^V pairings at the V vertices and projecting on the external tensor, so that $T_G(1)=1$. The factor $E!/|\mathrm{Aut}\,G|$ includes the sum over the labellings of the legs, which is legitimate at zero external momenta [15]. The integrals do not depend on N: since every graph carries its polynomial $T_G(N)$, one run gives the coefficients for all N.

2. *Constructing the integrands*

Every polynomial in Eq. (12) is the Kirchhoff polynomial of a minor of G, possibly with two vertices identified, and factorizes into the Kirchhoff polynomials of its blocks, which we call atoms. Expanding all forest terms into products of half-integer powers of atoms, with binomial coefficients that are dyadic rationals and hence exact in binary floating point, and adding the coefficients of identical products, removes every combination of forest terms that cancels identically; we call this step the pruning. Each graph becomes

$$f_G(x) = \sum_k c_k\, x_{\mathrm{ins},k} \prod_a P_a(x)^{p_{ka}/2}, \tag{13}$$

with atoms P_a, integer p_{ka} and exact c_k; terms with the same atoms, which differ only by the insertions, are grouped. The result of the expansion is a small program: the list of atoms (each a minor of G, evaluated at run time by a spanning-tree pass and a factorization of its loop matrix), the list of terms (coefficient, insertion monomial, atom exponents), and the tables of the sampling density of Sec. II C 2. One generic kernel evaluates any such program (Sec. II C 6).

The pruning removes up to two thirds of the raw terms. At six loops a four-point graph has on average 250 forests, which leave 116 terms in 22 groups with 19 distinct atoms (at most 55); at seven loops, in a sample of 300 four-point graphs, 583 forests, 219 terms, 39 groups and 26 atoms (at most 6128, 766, 126 and 85). The largest programs are those of two-point graphs with many nested two-point parts: at nine loops the chain of eight bubbles of C_9 has 3.4×10^5 forests and 2328 atoms, against at most 269 atoms for H_8 and 203 for B_8. The expansion is done once per graph on the host, in parallel over the graphs; its cost grows faster than the number of graphs (Table XVI: about 0.16 core-seconds per graph for B_8, 1.2 for H_9, 4.4 for C_9, and from 7.8 for B_{10} to 66 for C_{10} at ten loops), so that at ten loops it dominates a job. The built integrands of a list are therefore written once to a cache file, by a CPU-only job, and loaded by every subsequent run; a run with and without the cache gives bitwise the same results [check I4].

Graphs that vanish identically come out of the pruning with no terms at all, without any structural rule: graphs with a cut vertex (one-vertex-reducible graphs), which factorize at zero momentum [15] into pieces of which one is a renormalization part at exactly zero momentum, cancelled by its own subtraction; graphs with self-loops, for $L\geq2$; and, for H_L, generalized tadpoles, graphs with both external legs on one vertex, for which $W_G=0$. At three to seven loops these are 21–30% of the four-point and 15–17% of the two-point lists (more for H_L); at eight loops 18% and 14%, and at nine loops 12% of the two-point list. In production runs the one-vertex-reducible graphs of B_L and H_L are dropped before the expansion; their vanishing, and that of the graphs with self-loops, was verified explicitly up to five loops [check I2]. Those of C_L are expanded in every run, and the pruning leaves no term in any of them, at every order (independently verified up to seven loops [check I2]).

a. Twins. At zero momenta and equal masses, Eq. (8) depends on G only through its cycle matroid, so that 2-isomorphic graphs in Whitney's sense [57], which

we call twins, have equal integrals in any representation [15]: the 3986 seven-loop four-point graphs have 3018 distinct unsubtracted integrals, and the 3150 non-vanishing ones fall into 2391 classes. We checked that the subtracted integrands of twins, summed over the positions of the insertion, agree pointwise under the matroid map of the lines (all 3 pairs at four loops and all 25 non-vanishing pairs at five loops for B_L) [check I3]. Twins have different $O(N)$ factors, so that a merged class carries the sum of their weights. Twins were not merged in the runs of Sec. III: every class is one graph.

3. Consistency checks

The graphs, their weights and their integrands are checked, before any sampling, against results that are exact.

1. *Zero-dimensional sum rules.* In zero dimensions every integral equals one, so that $\sum_G S_G$ over a complete list is the coefficient of g^L in the corresponding vertex function of the zero-dimensional $O(N)$ model, known exactly from its moments (Appendix C). Summed over all graphs, one-vertex-reducible ones included, this tests the completeness of a list and its weights without any integral; for lists without self-loops the bare mass is shifted so that every self-loop cancels. Every list passes it exactly, with the full $O(N)$ polynomial, through ten loops (Table XVII) [check G2].

2. *Independent lists.* Converted to Nickel indices, our lists equal those of Ref. [14] (generated with FEYNGEN [58]) as sets, for $\Gamma^{(4)}$ at $L \leq 7$ and $\Gamma^{(2)}$ at $L \leq 8$ [check G3], and those of a brute-force generator in our code where that is practical ($\Gamma^{(4)}$ to five loops, $\Gamma^{(2)}$ to six). Two independent canonicalizers (one in the generator's string format, one on Nickel indices) agree on every graph, and all graphs of every list are distinct [check G1].

3. *Weights.* The weights and $O(N)$ polynomials equal those of the independent implementation of Refs. [15, 20] on all 260 graphs with $L \leq 5$ (its 251 and the nine graphs with self-loops its lists lacked) and on all 47 graphs of its four-loop tables [check G4].

4. *Integrands.* The expansion (13) agrees pointwise with an independent evaluation of the forest formula in 100-digit arithmetic (Sec. II E) [check I1]; vanishing graphs vanish [check I2], twins agree pointwise [check I3], and the exact two-loop sums $B_2 = 1/2$, $H_2 = 1/81$ and $C_2 = 1/12$ are reproduced to a few 10^{-7} by the sampled integrals (Sec. III A) [check N3].

5. *Large order.* Each new coefficient is compared with its leading large-order estimate (Sec. II E), a test of the complete chain that catches an error of a few per cent (Sec. II E) [check S3].

C. Numerical approach

1. Choosing the method

The earlier evaluations integrated each graph separately in momentum space, with one-loop subgraphs replaced by analytic effective vertices, on deterministic meshes [17], by adaptive cubature [15, 20], or with VEGAS [14, 21]. Because of the cancellations of the Z-factor route, and of the precision lost in finding the zero of β and in the resummation, eight to ten digits per graph were considered necessary [15]; high-precision methods based on integration-by-parts reduction [59, 60] are available for massive vacuum integrals only to five loops [61]. The cost of that strategy is set by its hardest graphs: every graph must reach many digits, and both the number of graphs and their dimension grow with the order, so that each new order costs more than the last by a large and growing factor.

We made the opposite choice deliberately: a method whose precision depends as little as possible on the order. Every integrand, whatever its size, is sampled by the same algorithm. Like any Monte Carlo estimate, an order sum then reaches a relative error ϵ in a time proportional to $1/\epsilon^2$; what the design controls is the constant of proportionality, and it aims at one that grows as slowly as possible with the order. Three choices serve that aim. The samples are shared among the graphs of an order in proportion to their spread (Sec. II C 3), so that the constant is set by an average over the graphs of the relative spread and the cost of one sample, and by a factor that measures the cancellations between graphs of opposite sign: the number of graphs does not enter it by itself. Each graph is sampled with a density that follows its singularities (Sec. II C 2), so that the weights stay bounded at every order. And each term of the forest formula (12) costs only a polynomial in L to evaluate, since its polynomials are determinants of $L \times L$ matrices (Kirchhoff's matrix-tree theorem). At low orders such a method cannot match the per-graph treatment, where effective vertices reduce each integral to a few dimensions and deterministic integration reaches eight to ten digits. At high orders, where the dimension grows with the number of lines and deterministic rules slow down accordingly, it should become far cheaper. How slowly its constant actually grows could only be measured: empirically, the time to a fixed relative precision grows by a factor of about 2.4–2.9 per loop (Sec. III B). The construction of the integrands, done once per graph, still grows with their number, that is factorially. It is a one-off cost on ordinary CPUs, but at high enough orders it becomes the dominant one: at the tenth order it already exceeds the sampling (Sec. III B). A precision that hardly depends on the order also keeps the statistical errors of

the highest coefficients, which carry the information on the large-order behaviour, from limiting the resummation; at the present orders they are already negligible there (Sec. II D 3).

The lower-order terms that enter a new coefficient together with its new order sums are known far more precisely and do not cancel them, so that the relative error of a coefficient is that of its order sums. What matters is therefore the relative error of each order sum, which no cancellation amplifies. Monte Carlo integration then becomes adequate: it requires no choice of variables per graph, it gives reliable error estimates when the weights are bounded, and it is naturally parallel. The parametric integrands, however, are singular on every face of Δ on which the parameters of a subgraph vanish, and the faces overlap at the corners; under uniform sampling the variance is already infinite on the face of a one-loop bubble. Sector decomposition [62, 63] removes these singularities, but in our tests it required 10^5–3×10^5 sectors for a six-loop graph and $(1\text{–}5)\times10^6$ for a seven-loop four-point graph, and lattice rules on the sector integrands [64] remained pre-asymptotic up to 10^9 points at 9–11 dimensions. We use instead tropical sampling [22, 52], a recent technique, increasingly used for Feynman integrals [22, 54, 65], which samples the Hepp sectors [45] rather than enumerating them. A sampler of the same kind, with an exponent for every subset of lines and the sector drawn by a recursion over the subsets, was used for the forest-subtracted parametric integrands of the electron anomalous magnetic moment, on CPUs and then on GPUs [51, 66, 67]; our exponents follow the forest terms of the zero-momentum subtraction (Sec. II C 2). Each graph is sampled with its own density. Sampling a whole order with one density [54] brings no gain for the unsubtracted integrands, which are all positive: in our tests at four and five loops a one-sample global sampler came within 3–4% of the optimally stratified per-graph sampling described below. Its advantage lies elsewhere: it draws graphs and points together, without enumerating the graphs, in polynomial time and memory, and it has been applied to finite correlators of a massive super-renormalizable theory in three dimensions [54].

2. *Forest-aware tropical sampling*

In the Hepp sector $x_{\sigma(1)}<\cdots<x_{\sigma(\ell)}$, with $S_j=\{\sigma(1),\dots,\sigma(j)\}$ and $t_j=x_{\sigma(j)}/x_{\sigma(j+1)}$, the density with respect to the projective measure $\prod_j dt_j/t_j$ is

$$p(x)=\frac{1}{J(E)}\prod_{j=1}^{\ell-1}t_j^{\,\omega(S_j)},$$
$$J(S)=\sum_{e\in S}\frac{J(S\setminus e)}{\omega(S\setminus e)},\qquad J(\{e\})=1, \tag{14}$$

where $J(E)$, a generalized Hepp bound [53], sums $\prod_j\omega(S_j)^{-1}$ over all sectors, and the exponents $\omega(S)$ follow the integrand. When the parameters of a set S of lines go to zero together, $x_e\to\lambda x_e$ for $e\in S$, an atom behaves as $\lambda^{h_a(S)}$, where $h_a(S)$ is the loop number of the lines of S that belong to it, and the term k of Eq. (13) as $\lambda^{\deg_S k}$, with $\deg_S k=\sum_a p_{ka}\,h_a(S)/2+n_k(S)$ and $n_k(S)$ the number of its insertions on lines of S. The term converges on this face if $|S|+\deg_S k>0$. We take

$$\omega(S)=s\Big[\,|S|+\min_k\deg_S k+\delta(S)\Big], \tag{15}$$

with the minimum over all terms that survive the pruning, $\delta(S)=1$ where $|S|+\min_k\deg_S k\le 0$ and zero elsewhere, and $s\in(0,1]$ a softening factor: $s=1$ gives the tropical density, and $s<1$ lowers every exponent by the same fraction, which tames the tails of the weights at the price of more samples in the bulk; we use $s=0.7$, chosen as described in point (iii) below. Both ω and J are tabulated on the host for all 2^ℓ subsets of lines, so that the cost grows as 2^ℓ rather than as the number $\ell!$ of sectors; the two tables take 1 MB per graph at $\ell=16$. A point is drawn by removing, from $S=E$ downwards, the largest line e of S with probability $J(S\setminus e)/[\omega(S\setminus e)\,J(S)]$, and setting the ratio of the largest remaining parameter to x_e to $t=v^{1/\omega(S\setminus e)}$ with v uniform [52]; a sample takes $2(\ell-1)$ uniform numbers. Its weight $w=f_G(x)\prod_e x_e/p(x)$, with f_G in projective form, has expectation $\int_\Delta d\mu\,f_G$, and the estimator is unbiased for any positive ω.

Three features of Eq. (15) adapt the tropical density to the subtracted integrands. (i) The minimum runs over all forest terms: a counterterm brings in the faces of its cograph, and a density built from the unsubtracted graph alone produced weights up to 8×10^4 times the mean. (ii) On faces where every term diverges, $|S|+\min_k\deg_S k\le 0$, the sum converges only through a t^2 subtraction, which raises the true exponent by two units (for the sunset, $|S|+\deg_S$ goes from 0 to 2, the measure from $\lambda^{-1}d\lambda$ to $\lambda^{+1}d\lambda$) and which the term-by-term minimum does not see; the shift $\delta=1$ keeps the density integrable and below the true behaviour, as we checked in extended precision on every shifted face of 280 integrands at four and five loops (no violation); a shift by two units changes the per-sample relative standard deviation of the order sum [ρ of Eq. (16) below] by less than 0.5%. On no face is the exponent negative before the shift, for any integrand up to ten loops: the builder refuses such integrands, on which the double-precision evaluation of the expanded t^2 terms would have round-off of infinite variance, and it built every class of every order [check G5]. (iii) The tropical density falls off faster than the integrand before the asymptotic regime: the positive powers of atoms in the t^2 terms give large but bounded prefactors that it does not see, and it ignores the factor $(\sum_e x_e)^{-(L/2+1)}$, which plausibly explains why softening helps the unsubtracted integrands as well. With $s=1$ the weight distributions had tails with a Hill index [68] as low as 1.9 and five-loop error estimates were unreliable. Softening all exponents by $s<1$ makes the weights vanish on every face. The value $s=0.7$ is close to optimal

for every kind of integrand at four and five loops, where it halves the per-sample standard deviation of both the subtracted and the unsubtracted integrands (any positive set of exponents gives a valid density, which can be fitted to the integrand [69]; the optimum found there for the corresponding uniform factor is 0.72); the smallest tail index is then 3.87, below 4, so that the fourth moment, and with it the error of the error below, is only indicative. We have not repeated the scan of s and of the shift beyond five loops. At six and seven loops replica ensembles test the resulting error estimates directly (Sec. III A) [check N7], and at eight loops the independent runs of B_8 and H_8 do (Table VII); beyond that they rest on the within-run statistics alone.

3. *Estimators, error estimates and allocation*

Every class is sampled independently; a class is one graph, since twins (Sec. II B 2) were not merged. Its contribution to the order sum $X_L = \sum_G a_G I_G$, with $a_G = \Pi_L S_G$ and I_G the integral in Eq. (9) summed over the positions of the insertions, is estimated as $a_G \bar{w}_G$ with variance $a_G^2 \sigma_G^2 / n_G$, where $\bar{w}_G$ and σ_G are the mean and standard deviation of its n_G weights. For each class we also record the kurtosis $\mathcal{K}_G$ of the weights, from which the relative standard error of σ_G, $[(\mathcal{K}_G - 1)/4n_G]^{1/2}$, is propagated to that of the error of X_L (the error of the error); the largest weight relative to the mean; and the largest round-off scale of a single sample (Sec. II C 5). An order is characterized by

$$\rho = \frac{\sum_G |a_G|\, \sigma_G}{|X_L|}, \qquad \rho_\tau = \frac{\sum_G |a_G|\, \sigma_G \sqrt{\tau_G}}{|X_L|}, \tag{16}$$

the per-sample relative standard deviation under optimal allocation, such that n samples give a relative error $\rho/\sqrt{n}$, and its counterpart per unit of computing time, with τ_G the measured cost per sample.

After a pilot of 2^{14} to 2^{16} samples per class in the runs of Sec. III, the samples are allocated by Neyman's rule [70], $n_G \propto |a_G| \sigma_G$, or, in a time-boxed run, $n_G \propto |a_G| \sigma_G / \sqrt{\tau_G}$ with τ_G in core or device seconds, which minimizes the error at fixed computing time. The main phase runs in rounds (eight by default), each of which brings every class to a fixed fraction of its allocation; the complete state is written after every round, and a run that reaches its time limit stops at the end of a unit of work. Since the samples of a class always form a prefix of its random-number stream, an interrupted run is identical to a shorter one. At $s = 0.7$ the pulls of single runs were Gaussian over 340 replica runs on CPUs at two to four loops, with no single run beyond three standard deviations, and over 400 replicas at three and four loops with the final code the mean pulls were within ± 0.1 of zero and their spread between 0.91 and 1.06; at $s = 1$ fatter tails had been seen. Since the sample variance of a heavy-tailed class converges slowly, we nevertheless judge agreement over independent seeds rather than from the error estimate of a single run (Sec. II E).

4. *Pooling the runs: the ledger*

Each class draws its random numbers from its own stream of a counter-based generator (Sec. II C 5), whose key is derived from the seed and from the class (kind, loop order, Nickel index). Two runs with different seeds therefore sample every class independently, whatever their hardware, allocation or duration and whenever they are contributed, and their per-class statistics pool exactly: with n_i samples, weight mean m_i and weight variance s_i^2 from run i, the pooled mean is $\sum_i n_i m_i / \sum_i n_i$ and the pooled sum of squares is $\sum_i [(n_i - 1) s_i^2 + n_i m_i^2]$. The ledger keeps, for every class, the registers of all its samples pooled: their number, mean and central moments up to the fourth, updated run by run with the pairwise formulas of Welford, Chan, Golub and LeVeque, and Pébay [71–73], together with the extremes of the weights, the round-off indicators and the time per sample, so that the ledger's size does not grow with the number of runs. For every run it records the seed, the code version and build signature, the device self-tests, the classes covered and the run's own statistics; every run file is kept as received in an archive outside the ledger, from which the ledger can be rebuilt bit for bit [check N5]. The per-order sums, the coefficients and their uncertainties, and the tables of this paper are regenerated from the ledger [check S2]. A run is admitted to the ledger only if its seed is new for that order, its build is a known one whose devices passed the start-up self-test against the host, it has no non-finite samples, and its classes are statistically consistent with the pooled ones (a χ^2 over the classes it shares with the ledger, which detects a wrong build or a failing device) [check N5]. The ledger and the tools are public with the code; a refinement is one command that chooses the order with the largest relative uncertainty, allocates the time across its classes by their measured cost and spread, runs with a fresh seed and submits its contribution. The precision of every coefficient can thus be improved indefinitely, by anyone, without recomputing what exists.

5. *Random numbers and arithmetic*

The random numbers come from the counter-based generator Philox4x32-10 [74], a pure function of a key and a counter; our implementation reproduces its published known-answer vectors [check N1]. Each class has its own key, a hash of the seed and of a string made of the kind of integrand, the loop order and the Nickel index, and the counter is the sample index. The random numbers of a class therefore depend only on the seed, the kind, L and the graph, not on its position in a list, on how an order is split into runs, or on the hardware. Runs at differ-

ent loop orders are independent even with the same seed (the largest measured correlation between index-matched classes, 5.4×10^{-3}, is consistent with zero), and replicas differ by the seed.

The evaluation is entirely in double precision (FP64), in logarithmic form. For each atom, the leading monomial at the sampled point is the product of the parameters of the chords of a minimum spanning tree, found by Kruskal's algorithm with the lines in increasing order of x_e; the loop matrix of the corresponding fundamental cycles, rescaled by the chord parameters, has entries that are ratios of parameters not larger than one and a determinant between one and the number of spanning trees, computed by an LDL^{T} factorization. Hence $\log P_a$ is obtained without overflow or underflow at any point the sampler can produce, however close to a face. The terms of Eq. (13) are added with Neumaier's compensated summation [75], and the sums and moments of the weights are accumulated with compensation as well. Pruning removes the terms that cancel identically; what remains are the cancellations of the subtractions close to the faces they protect, where double precision loses all relative accuracy pointwise (within about 3×10^{-7} of a sunset face). The absolute round-off of a sample remains of the order of the machine precision times $\sum_k |\text{terms}|$, which is negligible against σ_G: double and 80-bit extended precision at the same points differ by at most 1.7×10^{-8} standard deviations (rms) at four and five loops, and every run reports the worst single-sample ratio of this scale to σ_G, with a warning above 10^{-3}. Single precision is not used, as it is not adequate for subtracted integrands: already for the sunset-subtracted five-loop integrals of the intermediate scheme its error reaches 10^{-2} standard deviations per sample.

6. *A single kernel for CPUs and GPUs*

The integrator is written in C++. The sampler and the evaluation of Eq. (13) are written once, as header code compiled both by the host C++ compiler and by the CUDA compiler; the expansion of Sec. II A 3 runs on the host and produces, for each class, a flat image of its atoms, terms and tables. The unit of computation is a chunk of 4096 consecutive samples of one class, evaluated by a block of 128 lanes of 32 samples each. Each lane accumulates in registers with compensated summation, the block reduces its lanes along a fixed binary tree, chunks are merged in index order within units of 16 chunks, and units in order. Compensated merges are not associative bit for bit, so this order is part of the definition of the result: the accumulators of a class depend only on its key and on its number of samples. On CPUs they are bitwise identical for any number of threads; an emulator that runs the device program on the host, with the data layout of the device, reproduces them bitwise for any number of devices and blocks, any launch size and any order of execution. The kernel body is written against an executor that either loops over the lanes (CPU) or maps them onto threads (GPU), in phases that end with a barrier by construction; running the lanes in reverse order gives the same bits, which excludes a missing barrier. A GPU can therefore differ from the CPU only through the device mathematical library and the contraction of multiplications and additions into fused operations (the host code is compiled without contraction).

This is tested on every machine by golden runs: for 13 configurations (subtracted integrands of all three kinds, and intermediate-scheme four-point integrands, at three to six loops), reference files computed on a CPU with fixed seeds and 65 536 samples per class must be reproduced on the GPUs with identical sample counts and per-class means within 100 units of the round-off of the per-sample terms and within 10^{-4} standard deviations, on one device, on each device separately and with the work forcibly split among all of them [check N2]. On the CPU, perturbing every exponential and logarithm by up to 2 units in the last place moves the per-class means by less than one such unit. Before production each GPU also passes a self-test: device exponential and logarithm within 2 units in the last place of the host, exact square root and fused multiply-add, the Philox known answers, a signature of all compile-time limits and data layouts, 4096 samples of each of up to three classes compared one by one with the host (same Hepp sector, same weight to 10^{-11} of the sum of the absolute values of the terms), and one chunk through the production kernel.

On a GPU the kernel runs as persistent blocks that take chunks from an atomic counter, one class per block at a time, so that the threads of a warp follow the same integrand and chunks of classes with different costs balance automatically. Every GPU is driven by its own host thread, which takes units of 65 536 samples from a common queue in class order and keeps the images of the classes it uses in device memory, with least-recently-used eviction. Launches hold about half a second of work but at least two full waves of the grid; their measured durations are apportioned to the classes by their cycle counts and give the costs τ_G of the time-based allocation. The sample count of every unit is verified, uploads are read back, and every wait on a device is guarded by a watchdog; a device that fails is removed and its units are requeued on the others. Since a unit's result depends only on its counters, the result does not depend on how the work is split among the devices. The kernel's limits are compile-time constants, so the larger graphs need larger builds of the same sources: the default one handles up to 16 lines and 768 atoms per class, enough through eight loops; a variant with up to 18 lines and 8192 atoms covers nine loops (17 lines for $\Gamma^{(2)}$, 18 for $\Gamma^{(4)}$), and one with up to 20 lines, subgraphs of up to 11 loops and 16 384 atoms covers ten. The sector table of a class has 2^{ℓ} entries, about a million at ten loops. Each build has its own signature, and the ledger accepts runs only from known ones (Sec. II C 4).

D. Resummation

The critical exponents follow from the RG functions at the non-trivial zero $\bar{u}^*$ of β: $\eta = \eta(\bar{u}^*)$, $\nu = \nu(\bar{u}^*)$ and $\omega = \beta'(\bar{u}^*)$. The series are asymptotic, with coefficients that grow like $k!$ at large order k [5, 6], and $\bar{u}^*$ is of order one, so they must be resummed. The eighth-order series of Ref. [14] were not resummed there ("We postpone such study to a future work"), and to our knowledge they have not been resummed since; the standard fixed-dimension estimates rest on the shorter series analysed in Refs. [7–9]. We resum the eighth-order series, from Ref. [14] and from our own coefficients, and our ninth- and tenth-order series with the same method and the same settings, so that the effect of the new terms, and that of the more precise eighth-order coefficient of η, can be separated from that of the method.

1. Borel–Leroy transform and conformal mapping

We follow Refs. [7–9] (see also Ref. [2]). For a series $F(\bar{u}) = \sum_k F_k\, \bar{u}^k$ the Borel–Leroy transform

$$F(\bar{u}) = \int_0^\infty dt\; t^b\, e^{-t}\, \mathcal{B}_F^{(b)}(\bar{u}t),$$
$$\mathcal{B}_F^{(b)}(x) = \sum_k \frac{F_k}{\Gamma(k+b+1)}\, x^k, \tag{17}$$

has a convergent expansion, since at large order [2, 5, 6, 76]

$$F_k \simeq c_F\, k!\, (-a)^k\, k^{b_F} \big[1 + O(1/k)\big]. \tag{18}$$

The constant $a = 48\pi/[(N+8)A]$, fixed by the action $A = 113.3835$ of the instanton (Sec. II E), is the same for all RG functions ($a = 0.1477742$ at $N = 1$), while b_F and c_F depend on F: $b_F = 3 + N/2$ for β, for η_{ϕ^2} and hence for ν and $1/\nu$, and $b_F = 2 + N/2$ for η [2, 15]. Assuming that $\mathcal{B}_F^{(b)}$ is analytic in the plane cut along $(-\infty, -1/a]$, the mapping

$$w(x) = \frac{\sqrt{1+ax} - 1}{\sqrt{1+ax} + 1} \tag{19}$$

sends the cut plane onto the unit disk, and the series in $\bar{u}$ is replaced by

$$F(\bar{u}) \approx \sum_{k=0}^{K} d_k(\alpha, b) \int_0^\infty dt\; t^b\, e^{-t}\, \frac{[w(\bar{u}t)]^k}{[1 - w(\bar{u}t)]^\alpha}, \tag{20}$$

where the $d_k(\alpha, b)$ are fixed by matching the expansion of the right-hand side to $F_0, \ldots, F_K$. The factor $[1-w]^{-\alpha}$ makes the approximant of the Borel transform grow like $x^{\alpha/2}$ at large x (it is the parameter $\lambda = \alpha/2$ of Ref. [29]). Dividing F by $\bar{u}^s$ is equivalent to replacing (b, α) with $(b - s, \alpha + 2s)$, so the scan over α below contains the shifted series of Ref. [9], which also used a homographic transformation of the coupling and a factor $1 + r\bar{u}$ that we do not use.

2. Resummed functions, parameters and errors

The fixed point $\bar{u}^*$ is the zero of the resummed β, and ω is the slope of the resummed β there; η is resummed and evaluated at $\bar{u}^*$. For ν we resum the series of $1/\nu = 2 - \eta + \eta_{\phi^2}$ as a single series and take its inverse: the Callan–Symanzik relations (4) give $1/\nu$ directly, as a product of series, whereas the series of ν follows from it by an inversion, which turns any zero of $1/\nu$ in the complex $\bar{u}$ plane into a pole. As Ref. [9] summed both S and $1/S$ for each exponent, we also resum the series of ν itself as a second determination; the two differ by at most 1×10^{-3}, and the difference enters the error [check S2].

We scan $b = 0, \frac{1}{2}, \ldots, 12$ and $\alpha = -2, \ldots, 10$ in steps of $\frac{1}{2}$; the range of b contains the large-order values b_F, and beyond $\alpha \approx 10$ the successive truncations oscillate with growing amplitude (those of η from $\alpha \approx 12$; see Sec. III D). For each (b, α) we compute the results $F^{[K-2]}$, $F^{[K-1]}$ and $F^{[K]}$ of the last three truncations of Eq. (20) and, over the neighbourhood $|b' - b| \le 1$, $|\alpha' - \alpha| \le 1$, the largest change δ between successive truncations and half the range σ of $F^{[K]}$. As in Ref. [9], the central parameters (b^*, α^*) are those for which the result depends least on the truncation order and on the parameters, here the minimum of $\delta + \sigma$; taking the largest change over a neighbourhood rather than at the point itself discards the isolated points where successive truncations cross by accident, which are frequent at large α. The region of comparable stability is the set R of grid points where $\delta + \sigma$ is at most twice its minimum. The resummation error at fixed $\bar{u}^*$ is the largest of δ, σ and $\max_R |F^{[K]} - F^{[K]}(b^*, \alpha^*)|$ (for ν also the difference from the series of ν); as in Ref. [9], the change of the result when $\bar{u}^*$ moves by its own error is added to it. Every series is treated with the same grid and the same criteria. The truncation order is $K = 8$ for the eighth-order series, $K = 9$ for the ninth-order ones and, at tenth order, $K = 10$ for η and $1/\nu$ and $K = 11$ for β.

3. Propagating the Monte Carlo uncertainties

The coefficients carry statistical errors, which we propagate through the resummation and keep separate from its systematic error. Every per-order sum is stored class by class (Appendix E), with independent random numbers for every class and loop order (Sec. II C 5), and is linear in the class integrals: if these are drawn from Gaussians with their quoted errors, each sum at fixed N is a Gaussian variable with the propagated error of Sec. III C, independent of the other sums. We therefore draw the sums at each N (for the orders taken from Ref. [14], and for its own coefficients in the eighth-order row of Table XV, with their quoted errors), rebuild the coefficients through Eq. (4), and resum every draw with the central parameters; the standard deviation over 200 draws, with a fixed seed, is the statistical error. The sums at different N are correlated through the shared class inte-

grals, but no entry of Table XV combines different N. Drawing the class integrals themselves, for the ledger inputs of the eighth-order series at $N = 1$, gives the same spreads within 12%, as expected from the sampling error of 200 draws, and a linearized propagation agrees with the draws within 4% [check S2].

The statistical errors of our results are at most 5×10^{-5} for $\bar{u}^*$, 7×10^{-6} for ν, 3×10^{-6} for η and 1×10^{-4} for ω, at most 0.1 times the resummation errors (the largest ratios, for $\bar{u}^*$ at $N = 2$ and 3, come mostly from the coefficient of $\bar{u}^{11}$ in β). With the quoted errors of the eighth-order coefficients of Ref. [14], larger than ours, the ratio is at most 8×10^{-4} [check S2]. The resummed exponents are therefore limited by the resummation, not by the precision of the coefficients: more computing time spent on the present orders would not change them. Further progress needs more orders, which have so far reduced the errors only gradually (Table XV), and a better control of the non-analyticity at $\bar{u}^*$ (Sec. III D).

E. Validation strategy

Individual subtracted integrals have no external reference; only the order sums do. The checks are therefore organized as follows; their results are given in Sec. III.

a. Integrands. The expansion (13) agrees pointwise with an independent implementation in Python (to 1.4×10^{-14} of the sum of the absolute values of the terms, on 39 graphs at two to five loops) and with an independent multiprecision implementation with its own forests, Kirchhoff polynomials from the matrix-tree theorem and no pruning (all graphs at three to five loops and 20 graphs at six loops); pruned and unpruned expansions agree to the round-off of extended precision [check I1]. The vanishing graphs, the twins (Sec. II B) [checks I2, I3] and the shifted faces (Sec. II C 2) are checked as described above.

b. Exact values and known orders. The exact values of B_1, B_2, H_2, C_1 and C_2, and that of the subtracted integral of one three-loop graph (a sunset inserted in a ring), are reproduced at high statistics. The coefficients known from Ref. [14], B_L to $L = 7$ and H_L, C_L to $L = 8$, follow from its series through Eqs. (4) and (6), with errors propagated from those quoted there [check S1]; a run is compared with them through the pull

$$z = \frac{X - X_{\rm ref}}{\sqrt{\delta X^2 + \delta X_{\rm ref}^2}}, \tag{21}$$

where $X \pm \delta X$ is our value of an order sum and $X_{\rm ref} \pm \delta X_{\rm ref}$ the reference, at $N = 1$ and, by recombining the same integrals with their $O(N)$ polynomials, at $N = 0$, 2 and 3. The unsubtracted intermediate-scheme integrals, computed by the same code, are compared graph by graph with the values of Ref. [14] and with their order sums.

c. Error bars. Replica ensembles with independent seeds test the error estimates: the mean and spread of their pulls, and the scatter of every class among replicas against its quoted error (χ^2 per degree of freedom). The same per-class χ^2 over repeated runs tests classes with large kurtosis or large maximal weights individually. Since every class's integral is stored, two runs can be compared class by class, independently of any reference.

d. Hardware. Golden runs and device self-tests (Sec. II C 6) establish that the GPUs compute the same numbers as the CPU, up to the round-off of the device library, before any physics run is accepted [check N2].

e. New orders. For B_8–B_{10}, H_9, H_{10}, C_9 and C_{10}, which have no reference, and for H_8 and C_8, whose precision we improve, we rely on the following checks.

1. *A second route* (route B). The coefficients are also obtained from the intermediate-scheme integrals of $\Gamma^{(4)}(0)$, $\partial\Gamma^{(2)}/\partial p^2(0)$ and $\Gamma^{(2)}(0)$, in which only the sunset subgraphs are subtracted, through the Z factors and the reversion of the coupling series [15], as in Ref. [14]. In this scheme $\Gamma^{(2,1)}(0;0)$ follows from $\Gamma^{(2)}(0)$ through the exact identity [15]

$$\big(2-u_I\,\partial_{u_I}\big)\hat{\Gamma}_I^{(2)}(0) = \Big(2+\tfrac{N+2}{3}\,u_I-\tfrac{2(N+2)}{9}\,u_I^2\Big)\Gamma_I^{(2,1)}(0;0), \tag{22}$$

with $u_I = g_I/8\pi$, where $g_I = \lambda_0/m_I$ is the bare coupling in units of the mass m_I of the intermediate scheme, $\hat{\Gamma}_I^{(2)} = \Gamma_I^{(2)}/m_I^2$ and integrals in Nickel's normalization; the same identity checks intermediate-scheme data order by order. Route B shares with route A only the graph lists, the weights and the sampler, and, because of the cancellations of Sec. II A 1, it reaches a lower precision at equal cost. The formulas are derived to all orders and reproduce every published coefficient of Ref. [14] through eighth order from its per-graph values, at $N = 0$–4 [check S4]. We evaluated the intermediate-scheme integrals with eight to ten vertices on GPUs, in 26.8 GPU hours, and took those with fewer vertices from Ref. [14]; the ninth- and tenth-order coefficients of the two routes agree within 1.5 standard deviations (Sec. III C, Table XIV) [check S4]. Route B resolves them only to 7×10^{-4}–7×10^{-3} relative, and it has no check of the coefficient of $\bar{u}^{11}$ in β, which would need the four-point graphs with eleven vertices.

2. *Sum rules*: the eight- to ten-loop lists and weights pass the zero-dimensional sum rules (Sec. II B) [check G2].

3. *Large N*: the leading power of N of $B_L(N)$, $H_L(N)$ and $C_L(N)$ at each order is compared with the $1/N$ expansion in the same scheme [2, 15]. At $N = \infty$ with $\bar{u}$ fixed only the bubble chains survive, and $\beta = -\bar{u} + \bar{u}^2$, $\eta = 0$, $\eta_{\phi^2} = -\bar{u}$, $\nu = 1/(2-\bar{u})$: the terms in N^L of B_L, H_L and C_L must vanish for $L \geq 2$. No class of the ledger has an $O(N)$ weight of degree L, so they vanish identically, and in the

normalization of Eq. (7) (Sec. III C) $p_{\beta,k}$ has degree $k-2$, $p_{\eta,k}$ and $p_{\eta_{\phi^2},k}$ degree $k-1$, and the coefficient of N^k in $p_{\nu,k}$ is 2^{-k-1}. The terms in N^{L-1} follow from the $1/N$ corrections to Z, $\Gamma^{(4)}(0)$ and $\Gamma^{(2,1)}(0;0)$, one-dimensional integrals over the momentum of the bubble chain, which reproduce the leading coefficients of the polynomials of Ref. [14] through eighth order to 3×10^{-10} [check S5].

4. *Large order*: every coefficient is compared with its leading large-order estimate $c\,(-a)^k\,\Gamma(k+b+1)$, equivalent at large k to Eq. (18) [2, 5, 6, 76], with the instanton constants of $d=3$ [2, 6], $A=113.3835$ and the amplitudes c, so that in Nickel's coupling $\bar{u}$ the constant is $a=48\pi/[(N+8)A]=0.147774232\times9/(N+8)$ [Ref. [2], Eq. (28.73)], and $b=3+N/2$ for β and η_{ϕ^2}, $2+N/2$ for η, as implemented for the seven-loop computation [15]. The leading estimate alone catches an error of a few per cent in a new coefficient [check S3].

III. RESULTS

The numbers of Secs. III A and III B come from the runs recorded in the ledger (Sec. II C 4), on nodes with NVIDIA H200 GPUs: device self-tests, comparisons of the GPU and CPU code paths on identical random numbers, validation runs at two to five loops, replica ensembles, measurements of throughput, tuning and scaling, and complete orders at four to seven loops in production mode. Section III C gives the new coefficients, from eight to ten loops.

Unless stated otherwise, values refer to $N=1$, and B_L, H_L, C_L are the per-order sums of Eqs. (6) and (9). Errors are one standard deviation of the Monte Carlo estimate. At two loops we compare with the exact values of Eq. (A3), 1/2, 1/81 and 1/12 at $N=1$. From three loops on, we compare with reference values derived from the $O(N)$ series of β, η and ν of Ref. [14]: we invert the Callan–Symanzik relations (4) order by order [check S1] and propagate the quoted uncertainties of the series coefficients to first order, adding their effects in absolute value, which is conservative. Agreement is measured by the pull z of Eq. (21).

A. Validation on known orders

A computation of this kind can fail in three ways that a single run does not reveal: the integrands, or the code that evaluates them, can be wrong; the error bars can be too small; and the round-off of the cancellations inside the subtracted integrands can bias the means. Comparisons with known values test the first, but only as far as the error bars are right, and round-off at the level that matters lies below the precision of any reference. We therefore test the error bars and the round-off separately, without references, after establishing that runs on GPUs and on CPUs can be pooled.

a. GPU against CPU. On identical random numbers the GPU and CPU code paths evaluate the same points and can differ only through the device's mathematical library and the contraction of multiplications and additions (Sec. II C 6). Over the 13 configurations of the golden runs, 868 classes at three to six loops, the per-class means agree to within 0.59 units of the double-precision round-off scale of the per-sample terms, $\epsilon_{\mathrm{mach}}\sum|\mathrm{terms}|$, and the order sums to within 6×10^{-10} standard deviations, on each device alone and with the work split over all of them [check N2]; fused multiply-add contraction, which only the earliest runs used, stays within the same bound. The hardware therefore adds nothing to the error budget, and runs are pooled whatever device produced them.

b. Known values. The two-loop sums reproduce their exact values, $B_2=1/2$, $H_2=1/81$ and $C_2=1/12$ at $N=1$, within their errors and to relative uncertainties of a few 10^{-7} (Table III) [check N3].

All 15 comparisons at three to seven loops in Table III lie within 1.35 standard deviations, with $\chi^2=6.5$ for 15 values ($p=0.97$) [check N4]. The reference errors, added linearly over the coefficients of the N polynomials of Ref. [14], are conservative, and the pulls are correspondingly small. The complete seven-loop orders B_7 (β at $\bar{u}^8$, 3150 classes), H_7 (η at $\bar{u}^7$, 209) and C_7 (ν at $\bar{u}^7$, 265) reach relative uncertainties of 2.8×10^{-6}, 4.9×10^{-6} and 4.9×10^{-6}, below those of the references by factors of about 1.5, 11 and 2.9; at six loops, with the computing time we allocated, the precision of the references is matched (ratios of their uncertainty to ours of 0.82, 2.4 and 0.88). The four-loop references are known to 2×10^{-11} or better, which makes the four-loop sums the most stringent test of a bias common to all classes: with relative uncertainties of at most 7.6×10^{-7}, B_4, H_4 and C_4 deviate jointly from the references by $(-0.2\pm4.2)\times10^{-7}$, so any common bias is below 8.5×10^{-7} at 95% confidence, and that of H_4 alone below 1.5×10^{-6}.

Every class carries its $O(N)$ weight polynomial, so the same runs give the sums at any N, with errors that follow from the class errors without further assumptions. Recombined at $N=0$, 2, 3 and 4, the same 15 comparisons give χ^2 between 5.8 and 7.4 and no pull larger than 1.71 in magnitude [checks N4, S2]; they come from the same samples as those at $N=1$ and are correlated with them.

c. Error bars. The value of a class is the mean of its weights, and its error the standard deviation of the weights divided by the square root of their number (Sec. II C 3). The mean is unbiased whatever the distribution of the weights, but the error is only as reliable as the sample variance, and the subtractions make that the delicate part. Near a face of the parameter space where the lines of a renormalization part, a two- or four-point subgraph, shrink to zero, the subtracted integrand

TABLE III. Per-order sums at $N = 1$ compared with exact values ($L = 2$) and with values derived from the series of Ref. [14] ($L \geq 3$; uncertainties not shown are below 10^{-9} relative). The reference for C_7 was derived in the same way. This work: every contribution of the ledger (validation runs with a pilot of 65 536 samples per class, production-mode runs of complete orders with a pilot of 16 384 or 32 768, replicas; all with Neyman allocation by measured device time, within a fixed time box or up to a target uncertainty) pooled class by class. Classes: graphs with a non-vanishing subtracted integrand (twins were not merged). z: pull, Eq. (21). Cost: the integrand evaluations of all its runs, in Tf (1 Tf = 10^{12} function evaluations), a measure of the cost that does not depend on the hardware (Table V). Generated from the ledger [checks N3, N4, S2].

Sum	L	Reference	Classes	This work	$\delta X/\lvert X\rvert$	Cost [Tf]	z
B_L	2	1/2	1	0.50000006(33)	6.7×10^{-7}	1.9	+0.18
	3	0.598603706	6	0.59860364(29)	4.8×10^{-7}	18	−0.23
	4	0.967192687	19	0.96719250(71)	7.3×10^{-7}	23	−0.26
	5	1.89652536(49)	96	1.8965261(12)	6.2×10^{-7}	77	+0.57
	6	4.3006836(49)	502	4.3006914(59)	1.4×10^{-6}	33	+1.02
	7	10.917594(45)	3150	10.917628(31)	2.8×10^{-6}	17	+0.63
H_L	2	1/81	1	0.0123456808(24)	2.0×10^{-7}	2.7	+0.75
	3	−0.001542750058	1	−0.0015427487(12)	8.0×10^{-7}	34	+1.09
	4	0.004699104918	4	0.0046991056(34)	7.1×10^{-7}	15	+0.21
	5	0.002443913(10)	11	0.002443919(10)	4.2×10^{-6}	8	+0.37
	6	0.008020564(89)	50	0.008020530(36)	4.5×10^{-6}	4.3	−0.36
	7	0.01454955(80)	209	0.014550034(71)	4.9×10^{-6}	9.7	+0.60
C_L	2	1/12	1	0.083333395(56)	6.8×10^{-7}	1.5	+1.09
	3	0.07785997192	2	0.077860025(39)	5.0×10^{-7}	9.7	+1.35
	4	0.1006770275	5	0.100677024(76)	7.6×10^{-7}	24	−0.04
	5	0.170551919(82)	16	0.17055175(21)	1.2×10^{-6}	24	−0.77
	6	0.34591073(86)	62	0.34591043(98)	2.8×10^{-6}	9.8	−0.23
	7	0.801121(11)	265	0.8011143(39)	4.9×10^{-6}	6.5	−0.53

stays finite only through cancellations between its forest terms, and the tropical density, which keeps the variance finite (Sec. II C 2), follows it there only approximately. The weights of classes with renormalization parts therefore have heavy tails: at seven loops the kurtosis reaches 1.3×10^6 (a class of B_7) and the largest weight is up to 2.2×10^6 times the class mean (a class of H_7). With such tails the sample variance converges slowly, and the rare large weights that carry part of the variance may not yet have been drawn: in a short run the quoted error of such a class is more often too small than too large. Comparisons with references cannot detect this at the precision we reach, so we test the error bars directly, by repeating computations with independent seeds and comparing the scatter of the results with their quoted errors.

We made this test where it is most likely to fail, in short runs: eight independent replicas of each of B_6, H_6, C_6, B_7, H_7 and C_7, each sampling for four to eight minutes on one GPU, with medians of 10^7 to 10^9 samples per class. The replicas scatter as their quoted errors say: for the order sums $\chi^2 = 31.6$ for 42 degrees of freedom ($p = 0.88$), and class by class, over the 29 666 degrees of freedom of the classes sampled in every replica, $\chi^2/\mathrm{dof} = 1.003$ ($p = 0.36$); the classes that carry half of the variance of each order, the heavy-tailed ones among them, give χ^2/dof between 0.93 and 1.20 [check N7]. A second test needs no reference either, and grows with every run: each run merged into the ledger is compared, class by class, with the pool of the runs before it (Sec. II C 4). Over the 386 runs so compared, in 17 orders from 3 to 8 loops, the per-class χ^2 is 123 786 for 124 342 degrees of freedom ($\chi^2/\mathrm{dof} = 0.996$); the deviations of the runs' χ^2 from their expectations, in units of their standard deviations, have mean −0.07 and spread 0.99, as they should for correct errors; and the order sums on the common classes give $\chi^2 = 370.3$ for 386 runs ($p = 0.71$) [checks N5, S2].

Beyond seven loops the tails are heavier: the kurtosis reaches 1.6×10^7 (a class of H_{10}) and the largest weight 6.8×10^9 times the class mean (a class of C_{10}). At eight loops the independent runs of B_8 and H_8 scatter as their errors say (Table VII); the ten runs of B_7 scatter less than theirs, which suggests conservative rather than optimistic errors, and we quote the pooled errors without rescaling. For the single runs of C_8, H_9 and C_9, and for the slices of B_9, B_{10}, H_{10} and C_{10}, no replica test exists: their errors rest on the within-run statistics, and the runs driven to a target uncertainty stopped on their own error estimates, a data-dependent rule whose effect we have not quantified.

d. Round-off in the subtractions. The forest formula, Eq. (13), writes a subtracted integrand as a sum of terms, the bare integrand and one subtraction for each forest of renormalization parts, that diverge separately on the faces where the lines of a part shrink; only their

sum is integrable. Close to such a face the terms are large and nearly cancel, and the double-precision sum carries an absolute error of the order of $\epsilon_{\mathrm{mach}} \sum |\mathrm{terms}|$ (Sec. II C 5). Unlike the sampling error, this error does not decrease with more samples and is not part of the quoted errors; with a preferred sign it would bias the class means, below the precision of any reference. It grows with the number of renormalization parts, hence with the order, and with the number of samples, which come ever closer to the faces. Among the four classes of H_4, for instance, the only one with a two-point part, a sunset, has a single-sample round-off scale of up to 6×10^{-4} of its per-sample standard deviation, against about 10^{-13} in the other three.

Every run records, for each class, the largest round-off scale of a single sample and its root mean square over the samples, in units of the per-sample standard deviation. At seven loops the single-sample scale exceeds 10^{-3} in 144 of the 3150 classes of B_7 (largest 0.051), 54 of the 209 of H_7 and 76 of the 265 of C_7. The rms scale, which is what could shift a mean, stays far smaller: through seven loops it is at most 1.8×10^{-5} of the per-sample spread, and at eight to ten loops it exceeds 10^{-3} in 668 classes, up to 2.8 in a class of B_{10}. If every sample of every class were shifted coherently by its rms scale, the order sums would move by at most 0.17 of their statistical errors, at any order from four to ten loops (the largest for C_8).

Since the scales are estimates, we also measured the round-off itself where it is largest: the same 2×10^5 sample points of each of 55 classes, the 43 most exposed of B_8, H_8, C_8, H_9, C_9 and C_{10} and two controls per order at six and seven loops, were evaluated in double and in extended (64-bit-mantissa) precision on the CPU (B_9, B_{10} and H_{10}, completed later, are covered by the bound above). The measured rms round-off is typically 25 times below the scale (median), but in two classes of C_{10} it reaches 0.1 and 1.0 of the per-sample spread; there it adds to the sample variance, which the quoted errors include. No class shows a significant mean difference between the two precisions (the largest is 1.9 standard errors among the 55), and summed over the classes of an order the possible bias is at most 4×10^{-6} of the order's statistical error [check N6]. Since the GPU and CPU code paths agree on identical samples to a fraction of the round-off scale [check N2], the conclusion carries over to the GPU runs. No run produced a non-finite weight.

B. Performance

a. Hardware and kernel. The bulk of the numerical evaluations of this work were run on computing nodes with NVIDIA H200 GPUs and Intel Xeon Platinum 8568Y+ host CPUs. Each H200 GPU has compute capability 9.0, 132 streaming multiprocessors at 1.98 GHz, 140 GiB of memory with ECC enabled, and a double-precision rate of half the single-precision rate; the host CPUs also build the classes. The code was compiled with CUDA 12.1 for sm_90. The integration kernel uses 110 registers per thread and 1104 bytes of stack, so four blocks of 128 threads fit on a multiprocessor, and the grid has 528 blocks per device. Launches are sized to fill whole waves of the grid, with mean fill factors of 0.99–1.00. In the production-mode runs the devices were busy 96–99% of the wall time. The host builds the classes. For the 3150 classes of B_7 this takes 94 s summed over classes.

TABLE IV. Throughput with equal allocation over the classes of an order: samples per second summed over four NVIDIA H200 GPUs, and mean and maximum over the classes of the device time per sample on one GPU. b, h, c: the subtracted integrands of B_L, H_L, C_L. int4: the four-point integrands of the intermediate scheme of Ref. [14]. bare4: unsubtracted four-point integrands.

Kind	L	Classes	Rate [10^9/s]	ns/sample
b	4	19	2.42	1.65 / 2.25
	5	96	1.22	3.29 / 6.05
	6	502	0.652	6.13 / 15.5
h	4	4	2.00	2.00 / 2.51
	5	11	0.889	4.50 / 6.45
	6	50	0.431	9.25 / 17.0
c	4	5	0.997	4.01 / 5.16
	5	16	0.394	10.2 / 17.3
	6	62	0.162	24.5 / 54.3
int4	4	27	7.00	0.571 / 0.804
	5	129	4.58	0.874 / 1.61
	6	660	3.06	1.31 / 2.36
bare4	4	23	7.38	0.542 / 0.574
	5	105	4.92	0.812 / 0.844

b. Throughput. Table IV gives the sampling rates. For the subtracted integrands the cost per sample about doubles with every loop: 1.65, 3.29, 6.13 ns per sample and GPU for b at four, five and six loops, and 4.0, 10.2, 24.5 ns for c. At the same orders, a subtracted four-point integrand costs 2.9, 3.8 and 4.7 times as much per sample as the corresponding intermediate-scheme integrand. In the seven-loop production-mode runs, with Neyman allocation, the four GPUs drew 7.3×10^8, 3.1×10^8 and 1.1×10^8 samples per second for B_7, H_7 and C_7.

c. Tuning. The default kernel is limited by registers to a quarter of the maximum thread occupancy. Capping the registers so that six or eight blocks fit on a multiprocessor raises the throughput of the five-loop b and c kernels by 20–22% and 30–33%. This holds even though the six- and eight-block builds spill 96 and 232 bytes per thread to local memory. The deviations from the CPU do not change. The production-mode runs of this section used the default build, so their costs would drop by roughly a quarter if the five-loop gains carry over. At five loops, disabling FMA contraction (the default from version 0.4.0) costs 1% of throughput and brings the GPU

about five times closer to the CPU, which is compiled without contraction: the worst per-class deviation from the CPU on identical random numbers moves from 0.072 to 0.015.

d. Scaling. Classes are independent: each GPU draws units of work from a shared queue, and the host only merges per-class accumulators. On one, two and four GPUs the five-loop B_5 run drew 3.03, 6.01 and 12.1×10^8 samples per second of wall time, a parallel efficiency of 99% at two and at four GPUs. Beyond one node no communication is needed: the larger orders were split into slices of their lists, each run as an independent job on one or four GPUs and pooled in the ledger (Sec. III C).

e. Cost per decade of precision. For a Monte Carlo estimate the relative uncertainty after a device time T_{dev} (summed over GPUs) is $\delta X/|X| = \rho_\tau\, T_{\mathrm{dev}}^{-1/2}$, with ρ_τ of Eq. (16) under optimal allocation. Each decade of precision therefore costs a factor of 100 in time. We take the realized value $\rho_\tau = (\delta X/|X|)\sqrt{T_{\mathrm{dev}}}$ of each order, with $\delta X/|X|$ its uncertainty from all its runs pooled and T_{dev} the device time of those runs; for a single production-mode run it is within about 1% of the value predicted by Eq. (16). Table V and Fig. 1 give the device time $T_{\mathrm{dev}}(10^{-5}) = (\rho_\tau/10^{-5})^2$ needed for a relative uncertainty of 10^{-5}. From one order to the next, ρ_τ^2 grows by 4.1, 3.5, 3.1, 2.9, 2.4 and 2.9 for B_L (from four to ten loops) and by 6, 4.6, 3.8, 3.1, 2.6 and 2.9 for C_L (from four to ten): the growth factors fall from four to six at low orders to about three from seven loops on. For H_L the sequence (35, 1.1, 4.4, 2.6, 2.3 and 2.2) is irregular at low orders, because H_5 has an unusually large per-sample spread. At seven loops a relative uncertainty of 10^{-5} costs 1.6, 6.6 and 12 GPU hours for B_7, H_7 and C_7. Matching the precision of the reference value of B_6 (1.1×10^{-6}) would take 41 GPU hours, about 10 h on four GPUs.

f. Scaling with the loop order. Figure 1 shows that the two costs of an order grow in different ways. The CPU time of the class build curves upwards on the logarithmic scale: its growth per loop increases with the order, to a factor of 32 to 100 from seven loops on—the superexponential growth expected of a number of graphs that grows factorially times a sector table that doubles with every line. The GPU time of the sampling curves the other way: its growth per loop decreases with the order and settles, from seven loops on, at a factor of about 2.7 for B_L, 2.4 for H_L and 2.9 for C_L, so that the device time for a fixed relative uncertainty grows at most geometrically, not factorially as the number of graphs does (from seven to ten loops the lists grow by factors of 6.7 to 8.1 per loop for $\Gamma^{(4)}$ and 5.1 to 6.5 for $\Gamma^{(2)}$). Counted in integrand evaluations, which do not depend on the hardware, the growth is slower still: the evaluations needed for a relative uncertainty of 10^{-5} grow by a factor of 1.7–1.9 per loop (Table V), and the rest of the growth of the device time is the growing cost of one evaluation. Two effects combine. The cost of a sample grows slowly, because a class of ℓ lines costs a spanning-tree pass and an $L \times L$ factorization per atom and the number of atoms per class stays small after the pruning, 19 on average at six loops and 26 at seven (Sec. II B 2); and the relative variance of an order grows slowly, because the sampling density follows the forest-subtracted integrand sector by sector, so that the cancellations between forest terms happen inside each sample. Figure 2 shows the second effect graph by graph: under the Neyman allocation of Eq. (16) the device time of an order is $(\sum_G \kappa_G)^2/\epsilon^2$, with κ_G as in Fig. 2, and reconstructing it from the ledger reproduces the measured times to within 11%. The distribution of κ_G is broad, about one decade wide, and the effort is not concentrated in a few classes: at every order 54–62% of the classes are needed to account for 90% of the cost. The median κ_G falls by about 0.6 decades per loop, because the typical graph shrinks faster than the order sum grows, while the number of classes grows by a factor of 4.2 to 8.3, so that $\sum_G \kappa_G$ grows by 1.5 to 2.1 per loop and the cost by its square. At this rate the sums of the tenth order cost up to a few hundred GPU hours at 10^{-5}

TABLE V. Cost of the sums on NVIDIA H200 GPUs. $\delta X/|X|$: the relative uncertainty of the sum, all its runs pooled. $\rho_\tau = (\delta X/|X|)\sqrt{T_{\mathrm{dev}}}$, with T_{dev} the device time of those runs summed over their GPUs, does not depend on how long they ran; $T_{\mathrm{dev}}(10^{-5}) = \rho_\tau^2/10^{-10}$ is the device time for a relative uncertainty of 10^{-5}, in GPU hours, the quantity to compare between orders (Fig. 1). $\rho_n = (\delta X/|X|)\sqrt{n}$ and $n(10^{-5}) = \rho_n^2/10^{-10}$ are the same with n, the number of integrand evaluations of those runs, in place of the device time, in Tf (10^{12} evaluations): they do not depend on the hardware. The numbers of classes are in Tables III and VII. Generated from the ledger [check S2].

Sum	L	$\delta X/\lvert X\rvert$ [10^{-5}]	ρ_τ [$10^{-4}\,\mathrm{s}^{1/2}$]	$T_{\mathrm{dev}}(10^{-5})$ [h]	ρ_n	$n(10^{-5})$ [Tf]
B_L	4	0.07	1.15	0.037	3.47	0.12
	5	0.06	2.34	0.15	5.44	0.30
	6	0.14	4.35	0.52	7.89	0.62
	7	0.28	7.62	1.6	11.5	1.3
	8	0.25	12.9	4.6	15.6	2.4
	9	0.98	19.8	11	20.8	4.3
	10	1.15	33.7	32	28.6	8.2
H_L	4	0.07	1.17	0.038	2.80	0.079
	5	0.42	6.89	1.3	11.8	1.4
	6	0.45	7.34	1.5	9.36	0.88
	7	0.49	15.4	6.6	15.2	2.3
	8	0.44	24.7	17	19.7	3.9
	9	0.99	37.8	40	27.5	7.5
	10	1.42	56.3	88	34.2	12
C_L	4	0.08	2.05	0.12	3.72	0.14
	5	0.12	5.01	0.70	5.92	0.35
	6	0.28	10.8	3.2	8.83	0.78
	7	0.49	21.0	12	12.5	1.6
	8	0.99	36.9	38	17.4	3.0
	9	1.00	59.3	98	23.0	5.3
	10	2.89	102	288	32.1	10

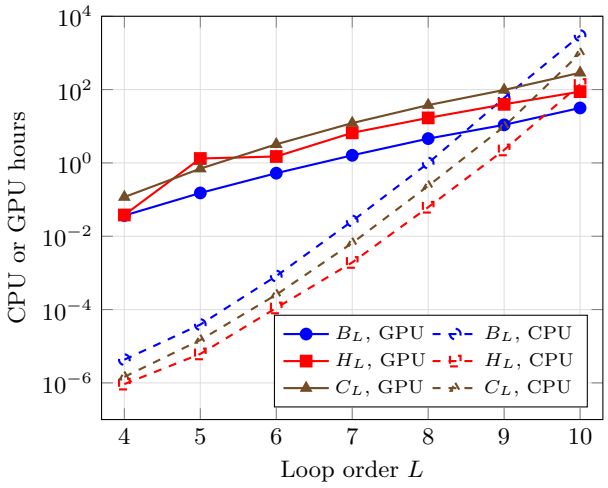


FIG. 1. The two costs of an order: the CPU time of the class build (dashed; the sum over the classes of the single-thread build times recorded in the ledger, in hours of one core of an Intel Xeon Platinum 8568Y+) and the GPU time of the sampling to a relative uncertainty of 10^{-5} (solid; the measured $T_{\mathrm{dev}}(10^{-5})$ of Table V, in hours of one NVIDIA H200 GPU, which scales as the inverse square of the target uncertainty). From seven loops on the sampling grows by a factor of 2.4–2.9 per loop, the build by 32 to 100, since the sector table of a class spans the 2^{ℓ} subsets of its lines and the minimum over its terms grows with them. For H_L and C_L the sampling dominates through nine loops and the build at ten; for B_L, whose classes are many and small, the build dominates from nine loops on. Generated from the ledger [check S2].

(from 11 for B_9 to 288 for C_{10}), and the limiting factor becomes the generation and the symbolic preparation of the graphs rather than the sampling. Figure 1 shows the crossing: the class build, which is dominated by the sector table over the 2^{ℓ} subsets of lines, took 9.8 CPU hours for C_9 against 98 GPU hours of sampling, but grows by a factor of 32 to 100 per loop and increasingly so with the order, so that at ten loops it exceeds the sampling and is run on CPU nodes, once per class, into the class cache of Sec. II B 2. For the three sums of the tenth order, B_9, H_{10} and C_{10}, the build took 1 200 core-hours, against 89 GPU hours of sampling to the precisions of Table VII; in node-hours, with 32 cores per CPU node and the eight H200 GPUs of a GPU node, that is 37 against 11.

g. Route A against route B. With the integrator of Sec. II C the per-sample relative standard deviation of the order sum, ρ of Eq. (16), is 3.38 and 5.22 for B_4 and B_5, and 2.79 and 11.6 for H_4 and H_5, against 19.2, 40.5, 13.8 and 101 for the intermediate-scheme input of route B alone at the same order: the subtracted integrands need 32, 60, 24 and 76 times fewer samples or, as each sample costs 2.4 to 3.9 times more on the CPU, 14, 22, 8 and 19 times less work at equal instruction count. These gains are lower bounds, since the route-B figures include only the highest-order input, and they grow with the loop order. On the GPUs, where a subtracted four-point sample costs 2.9 and 3.8 times more at four and five loops (Table IV), the GPU time for B_4 and B_5 drops by factors of about 11 and 16. At the ninth and tenth orders the route-B runs of Table XIV, 26.8 GPU hours in all, give coefficients whose uncertainties are 90 to 500 times those of route A.

h. Comparison with earlier computations. Table VI sets these times against those of the two complete computations of the fixed-dimension series [14, 17]. Three differences must be kept in mind. First, both computations produce individual graph values, from which the RG functions follow after the cancellations of the Z-factor route (route B). In that route the four-point input at four to six loops exceeds the net coefficient by factors of 29 to 74 (Sec. II A 1), so per-graph relative errors are not relative errors of the RG coefficients. Our uncertainties are those of the per-order sums themselves, from which the coefficients follow algebraically. Second, the errors are of different kinds: deterministic convergence estimates, VEGAS errors per graph, and statistical errors of an order sum. Third, the hardware spans five decades. What the comparison does show is that a complete order, with all its graphs and subtractions and with an uncertainty of 10^{-5}, now takes minutes to a day on one node of four GPUs through nine loops, and up to a few hundred GPU hours, split into independent jobs, at ten, without any graph-specific work. Whether our values are more precise is a separate question: at six loops, with the computing time we allocated, ours are about as precise as the existing ones or more, and at seven loops more precise, by factors of about 1.5, 11 and 2.9 for B_7, H_7 and C_7 (Table III).

C. New coefficients

Once the integration scheme had been validated on the known orders (Sec. III A), several production runs computed the four-point sums B_8–B_{10} and the two-point sums H_8–H_{10} and C_8–C_{10}, and repeated B_7, H_7 and C_7 at higher precision (Table VII). B_8 and H_8 were computed as independent replicas with fresh seeds, pooled in the ledger, with a pilot of 16 384 samples per class, allocation by measured device time and density parameter $s = 0.7$; H_9, C_9, C_8 and the repetitions of B_7, H_7 and C_7 were run in one queue driven to target relative uncertainties, 10^{-5} for H_9, C_9 and C_8, 3×10^{-6} for B_7 and 5×10^{-6} for H_7 and C_7, with checkpoints every 15 minutes from which a run resumes across sessions. The graph lists (Appendix B) contain 26 540 four-point graphs at eight loops, and 1622 and 9184 two-point graphs at eight and nine loops. Without the one-vertex-reducible graphs and, for H_L, the generalized tadpoles, 21 689 graphs remain for B_8, 1157 and 1387 for H_8 and C_8, and 6776 and 8038 for H_9 and C_9 (Table XVI); every one of them is a class, since twins were not merged. Their builds take 3 600, 220, 850, 8 100 and 35 000 core-seconds. H_8 reaches a relative uncertainty of 4.4×10^{-6}, so that the

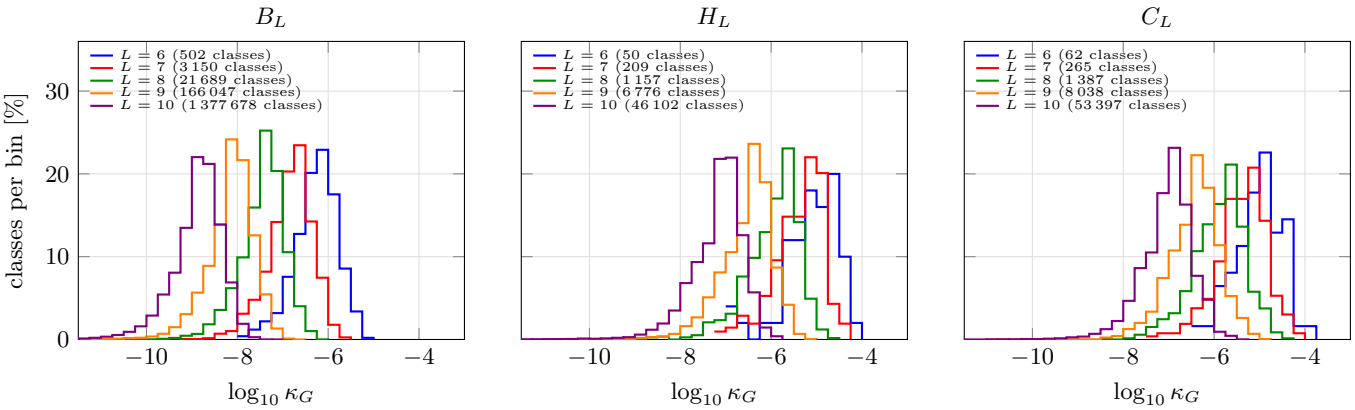


FIG. 2. Distribution over the classes of an order of the Neyman cost weight $\kappa_G = |a_G|\sigma_G\sqrt{\tau_G}/|X_L|$, where $a_G\sigma_G$ is the standard deviation per sample of the weighted contribution of class G, τ_G its device time per sample and X_L the order sum: the device time that brings X_L to the relative uncertainty ϵ is $T(\epsilon) = (\sum_G \kappa_G)^2/\epsilon^2$, and class G receives the fraction $\kappa_G/\sum_{G'} \kappa_{G'}$ of it [Eq. (16)]. Each histogram is normalized to the number of classes of its order (in the legends); the bins are a quarter of a decade wide, and κ_G is in $\sqrt{\text{s}}$ of H200 time, from the pooled per-class statistics. From one order to the next the distributions move to the left by about 0.6 decades while the number of classes grows by a factor of 4.2 to 8.3, so that $\sum_G \kappa_G$ grows only by a factor of 1.5 to 2.1 per loop. The resulting $T(10^{-5})$, 4.54, 37.1 and 94.6 GPU hours for B_8, H_9 and C_9, reproduces the measured times of Table V and Fig. 1 to within 11%. Generated from the ledger [check S2].

TABLE VI. Computing effort of complete fixed-dimension computations. The entries are not like for like: they differ in the quantities computed (individual graphs in Refs. [14, 17], complete per-order sums here), in the nature of the uncertainty (deterministic convergence estimates, VEGAS errors per graph, statistical errors per order), and in the hardware. This work: per loop order, the sums computed and their classes, the device time for a relative uncertainty of 10^{-5} of each sum (GPU h, Table V) and, from eight loops on, the CPU time of the class construction (CPU h). The row of this work is generated from the ledger [check S2].

Computation	Quantities	Hardware	Time	Uncertainty
Nickel et al. [13, 17]	$\Gamma^{(2)}$, $\Gamma^{(4)}$ to six loops, graph by graph	IBM 370/155, CDC 6600	$\approx$ 20 h on the IBM and $\approx$ 3 h on the CDC, for the whole evaluation	$\sim 10^{-8}$ absolute per integral
Phi4tools [14]	$\Gamma^{(4)}$ to seven loops, $\Gamma^{(2)}$ to eight, graph by graph	two CPU clusters	1–2 days for each integral of dimension ≥ 6	a few 10^{-5} ($\Gamma^{(2)}$, $\Gamma^{(4)}$), a few 10^{-4} ($\partial_{p^2}\Gamma^{(2)}$) per graph
This work, six to ten loops	B_6, H_6, C_6: 614 classes	NVIDIA H200 GPUs, Intel Xeon Platinum 8568Y+ CPUs	0.52, 1.5, 3.2 GPU h	10^{-5} relative, per order sum
	B_7, H_7, C_7: 3 624 classes		1.6, 6.6, 12 GPU h	
	B_8, H_8, C_8: 24 233 classes		4.6, 17, 38 GPU h; 1.3 CPU h	
	B_9, H_9, C_9: 180 861 classes		11, 40, 98 GPU h; 71 CPU h	
	B_{10}, H_{10}, C_{10}: 1 477 177 classes		32, 88, 288 GPU h; 4 100 CPU h	

coefficient of $\bar{u}^8$ in η is 120 times more precise than the value derived from Ref. [14], whose relative uncertainty, 5.4×10^{-4}, is the largest among the eighth-order coefficients (Table I). The improvement also matters for β: through the term $2\eta u$ of the Callan–Symanzik relation, the coefficient of $\bar{u}^9$ in β depends on H_8 as well as on B_8. The sums of the tenth order, B_9, H_{10} and C_{10}, with 166 047, 46 102 and 53 397 classes, and that of the eleventh, B_{10}, with 1 377 678, were too large for one node: each list was split into disjoint slices, each slice was built into a class cache on CPU nodes and then sampled by its own GPU job, and the slices were pooled in the ledger like any other runs. Since every class has its own random-number stream, the result does not depend on how the list was split or where the slices ran.

a. Series coefficients. Below the orders we compute anew, each order sum entering the polynomials $p_{X,k}(N)$ of Eq. (7) is the more precise of ours and the one derived from Ref. [14] (at present B_3–B_6, H_3, H_4 and C_3–C_6 from Ref. [14], the exact values at one and two loops, and all the others from the ledger); with our own sums at every order the coefficients change by at most 0.0075 of their

TABLE VII. Per-order sums at $N = 1$ from all runs pooled in the ledger. Enters: the function and the power of $\bar{u}$ whose coefficient the sum gives. Graphs: the number of graphs of the list (without self-loops, one-vertex-reducible ones included; Table XVI). Classes: the number of non-vanishing integrands, every one of them sampled. $\delta X_\mathrm{s}/|X|$ and χ^2/dof: where several seeds each cover the whole order, the χ^2 of their order sums about the inverse-variance mean, and the relative error of the pool rescaled by the scatter, $(\delta X/|X|)\sqrt{\chi^2/\mathrm{dof}}$; the per-seed values are in the ledger. Previous: derived from the series of Ref. [14] as in Table III. Cost: the integrand evaluations of all its runs, in Tf, as in Table III. The tenth- and eleventh-order sums were split into disjoint slices of their lists, run separately. Generated from the ledger [check S2].

Sum	Enters	Graphs	Classes	Previous	This work	$\delta X/\lvert X\rvert$	Cost [Tf]	$\delta X_\mathrm{s}/\lvert X\rvert$	χ^2/dof
B_7	β, $\bar{u}^8$	3986	3150	10.917594(45)	10.917628(31)	2.8×10^{-6}	17	1.4×10^{-6}	2.3/9
B_8	β, $\bar{u}^9$	26 540	21 689	—	30.469859(76)	2.5×10^{-6}	39	3.6×10^{-6}	4.3/2
B_9	β, $\bar{u}^{10}$	197 022	166 047	—	92.27709(91)	9.8×10^{-6}	4.5	—	—
B_{10}	β, $\bar{u}^{11}$	1 593 254	1 377 678	—	300.5548(35)	1.2×10^{-5}	6.2	—	—
H_7	η, $\bar{u}^7$	317	209	0.01454955(80)	0.014550034(71)	4.9×10^{-6}	9.7	4.6×10^{-6}	7.8/9
H_8	η, $\bar{u}^8$	1622	1157	0.037491(20)	0.03750774(17)	4.4×10^{-6}	20	4.5×10^{-6}	2.0/2
H_9	η, $\bar{u}^9$	9184	6776	—	0.09845268(98)	9.9×10^{-6}	7.7	—	—
H_{10}	η, $\bar{u}^{10}$	59 614	46 102	—	0.2879452(41)	1.4×10^{-5}	5.8	—	—
C_7	ν, $\bar{u}^7$	317	265	0.801121(11)	0.8011143(39)	4.9×10^{-6}	6.5	6.6×10^{-6}	16.2/9
C_8	ν, $\bar{u}^8$	1622	1387	2.07009(11)	2.070192(21)	9.9×10^{-6}	3.1	—	—
C_9	ν, $\bar{u}^9$	9184	8038	—	5.859068(58)	1.0×10^{-5}	5.3	—	—
C_{10}	ν, $\bar{u}^{10}$	59 614	53 397	—	17.95856(52)	2.9×10^{-5}	1.2	—	—

TABLE VIII. Coefficients a_j of the polynomials $p_{X,k}(N) = \sum_j a_j N^j$ of Eq. (7) for the new and improved terms of β and η. Powers beyond the degree of a polynomial are left empty. The coefficients are correlated; their covariance is in the Supplemental Material (`series/generated/polynomials.json`). Generated from the ledger [check S2].

j	$p_{\beta,9}$	$p_{\beta,10}$	$p_{\beta,11}$	$p_{\eta,8}$	$p_{\eta,9}$	$p_{\eta,10}$
0	$-6.432899(13) \times 10^7$	$1.130031(9) \times 10^9$	$-2.138589(19) \times 10^{10}$	58315.6(2)	$-8.88414(7) \times 10^5$	$1.499965(16) \times 10^7$
1	$-3.225258(10) \times 10^7$	$6.08032(7) \times 10^8$	$-1.224609(16) \times 10^{10}$	52164.0(2)	$-8.28881(8) \times 10^5$	$1.46114(2) \times 10^7$
2	$-5.02956(3) \times 10^6$	$1.07233(2) \times 10^8$	$-2.39487(6) \times 10^9$	13685.34(10)	$-2.36573(4) \times 10^5$	$4.54534(10) \times 10^6$
3	$-2.53755(5) \times 10^5$	$6.9100(5) \times 10^6$	$-1.86296(14) \times 10^8$	1113.67(2)	−22922.3(9)	$5.3144(3) \times 10^5$
4	−1755.7(7)	$1.1017(7) \times 10^5$	$-4.715(2) \times 10^6$	11.847(3)	−379.02(15)	18224(5)
5	0.56(6)	−392(8)	$-4.1(3) \times 10^3$	0.1561(3)	11.036(18)	−41.7(6)
6	0.331(5)	2.5(7)	$-1.1(3) \times 10^2$	−0.07039(3)	−0.3953(18)	4.15(8)
7	0.0076(2)	0.24(5)	2(2)	−0.0034718(13)	−0.07014(15)	−0.687(7)
8		0.0070(19)	0.18(16)		−0.002790(7)	−0.0673(6)
9			0.009(6)			−0.00228(2)

TABLE IX. As Table VIII, for η_{ϕ^2} and ν. The coefficient of N^k in $p_{\nu,k}$ is exactly 2^{-k-1}, and $p_{\eta_{\phi^2},k}$ has degree $k-1$, as the limit $N \to \infty$ requires [check S5]. Generated from the ledger [check S2].

j	$p_{\eta_{\phi^2},8}$	$p_{\eta_{\phi^2},9}$	$p_{\eta_{\phi^2},10}$	$p_{\nu,8}$	$p_{\nu,9}$	$p_{\nu,10}$
0	$3.10364(3) \times 10^6$	$-5.07683(4) \times 10^7$	$9.0136(2) \times 10^8$	$-6.52885(6) \times 10^5$	$1.085686(10) \times 10^7$	$-1.95287(5) \times 10^8$
1	$2.90212(3) \times 10^6$	$-4.95301(5) \times 10^7$	$9.1367(3) \times 10^8$	$-5.67424(7) \times 10^5$	$9.93775(12) \times 10^6$	$-1.87088(6) \times 10^8$
2	$8.33826(13) \times 10^5$	$-1.54661(2) \times 10^7$	$3.06457(14) \times 10^8$	$-1.39066(3) \times 10^5$	$2.72429(6) \times 10^6$	$-5.6224(3) \times 10^7$
3	82917(3)	$-1.82787(6) \times 10^6$	$4.1482(4) \times 10^7$	−9076.3(7)	$2.39743(16) \times 10^5$	$-6.0652(10) \times 10^6$
4	1734.1(4)	$-6.5507(10) \times 10^4$	$2.0285(8) \times 10^6$	115.87(10)	2450(3)	$-1.5181(19) \times 10^5$
5	−30.08(4)	28.1(13)	$1.349(11) \times 10^4$	9.238(11)	20.6(3)	$2.67(3) \times 10^3$
6	−1.154(4)	−22.46(13)	$-2.05(13) \times 10^2$	2.2339(9)	16.90(4)	49(3)
7	−0.02393(19)	−0.835(10)	−17.5(12)	0.11383(5)	1.941(3)	20.9(3)
8		−0.0169(5)	−0.64(9)	2^{-9}	0.07480(13)	1.58(2)
9			−0.012(4)		2^{-10}	0.0489(10)
10						2^{-11}

standard deviations. Tables VIII and IX give the new polynomials. Their coefficients are correlated, so we also give the coefficients evaluated at $N = 0$–4 directly from the classes (Tables XI, XII and XIII). The full covariance of the coefficients of all the polynomials, propagated linearly from the per-class integrals and from the errors quoted in Ref. [14], is in the Supplemental Material, as the file `series/generated/polynomials.json` and, one row per coefficient, `polynomials.tsv` (Appendix E). The uncertainties of the previous values in Tables XII and XIII are the errors quoted in Ref. [14] for the coefficients of the polynomials, added linearly.

At $N = 1$, with the quoted errors propagated through the Callan–Symanzik relations [check S2], the coefficient of $\bar{u}^8$ in η is 0.002910571(13), against 0.0029093(16) from Ref. [14], a difference of +0.8 of its standard deviation (+2.9 with the per-graph errors of Ref. [14] added in quadrature, see below), and that of $\bar{u}^8$ in ν is −0.0317869(4), against −0.0317852(17), a difference of −1.0; ours are 120 and 4 times more precise. The new coefficients are in Tables XI–XIII: for instance, the coefficients of $\bar{u}^9$, $\bar{u}^{10}$ and $\bar{u}^{11}$ in β are −2.366420(6), 4.78115(5) and −10.38718(12), and those of $\bar{u}^{10}$ in η and ν are 0.00995358(14) and −0.127571(5). The ratios $|\beta_{k+1}/\beta_k|$ of successive coefficients of β at $N = 1$ are 1.32, 1.51, 1.70, 1.86, 2.02 and 2.17 for $k = 5$–10. They grow as the large-order behaviour of Sec. II D requires: Eq. (18) has them approach $a(k + b + 1)$, which runs from 1.40 at $k = 5$ to 2.14 at $k = 10$ (Table X gives the quotients of the coefficients by their leading estimates).

b. Large-order and large-N tests. Against the leading large-order estimate of Sec. II E (Table X, $N = 1$) the signs agree from $k = 4$ on; the ratio coefficient/estimate is 1.025, 1.022, 1.030, 1.043 and 1.058 for β at $k = 7$ to 11, so the new orders lie within six per cent of the leading asymptote, with a shallow minimum at $k = 8$; for η_{ϕ^2} the ratio lies between 0.81 and 0.86 from $k = 6$ on, and for η, whose asymptotic regime sets in late, between 0.54 and 0.65 [check S3]. The ratios are not yet asymptotic, since all of them should tend to 1. The subleading corrections to Eq. (18) are powers of $1/k$, so we fitted the ratios of β and η_{ϕ^2}, at every N from 0 to 4, with $c_0(1 + d_1/k + d_2/k^2)$: the form reproduces them from $k = 6$ on, to 4×10^{-4} for β and 3×10^{-3} for η_{ϕ^2}, but with large coefficients, d_1 between −4.9 and −3.0 and d_2 between 13 and 17, nearly equal for β and η_{ϕ^2} at each N, and with a limit c_0 between 1.2 and 1.8 instead of 1: with corrections this large, orders this low cannot fix the limit, and the fit describes the known terms rather than extrapolating them. As a test of the new coefficients, though, the fit is much sharper than the leading estimate: predicted from the lower orders alone, the coefficients of β from $k = 9$ on agree with the computed ones to 1×10^{-3} at every N (4×10^{-4} at $N = 1$) and those of η_{ϕ^2} to 5×10^{-3}, against Monte Carlo errors of about 10^{-5}, so that an error of several times 1×10^{-3} in a new coefficient of β would stand out [checks S2, S3]. The leading coefficients of the 27 order sums of the ledger, $L = 2$–10, agree with the $1/N$ expansion (Sec. II E) within their errors: $\chi^2 = 23.7$ for 27 degrees of freedom ($p = 0.65$), with no pull above 2.3 [check S5]. At the highest orders the leading coefficients carry relative errors of tens of per cent (Tables VIII and IX), and the test resolves them only at that level.

c. Two independent routes. As a check on the subtraction and on the Callan–Symanzik relations, we also obtained the ninth- and tenth-order coefficients along route B (Sec. II E), from the intermediate-scheme vertex functions through the Z factors and the reversion of the coupling series, as in Refs. [14, 17]. The graph lists, the weights and the sampler are shared, but the integrands are the intermediate-scheme ones, and the route shares no subtraction with route A. We sampled the four-

TABLE X. Large-order test at $N = 1$: the coefficient of $\bar{u}^k$ in β, η and η_{ϕ^2} divided by its leading large-order estimate (Sec. II E). Generated from the ledger [checks S2, S3].

k	β	η	η_{ϕ^2}
3	1.751	-0.548	1.165
4	1.313	1.120	0.937
5	1.121	0.368	0.838
6	1.051	0.622	0.812
7	1.025	0.542	0.808
8	1.022	0.598	0.821
9	1.030	0.617	0.839
10	1.043	0.652	0.860
11	1.058	—	—

TABLE XI. Coefficients β_k of $\bar{u}^k$ in β [Eq. (7)] for $N = 0$–4; β_{10} follows from B_9 and H_9, β_{11} from B_{10} and H_{10}. Generated from the ledger [check S2].

N	β_9	β_{10}	β_{11}
0	−3.834306(8)	8.41939(6)	−19.91716(17)
1	−2.366420(6)	4.78115(5)	−10.38718(12)
2	−1.510105(5)	2.83206(4)	−5.70235(9)
3	−0.995275(4)	1.74500(3)	−3.28063(7)
4	−0.675610(3)	1.11399(2)	−1.96751(5)

TABLE XII. Coefficients η_k of $\bar{u}^k$ in η for $N = 0$–4, in units of 10^{-3}: η_8 from the polynomials of Ref. [14] (their quoted errors added linearly) and from this work, and the new η_9 and η_{10} (from H_{10}). Generated from the ledger [check S2].

N	$10^3\eta_8$ [14]	$10^3\eta_8$	$10^3\eta_9$	$10^3\eta_{10}$
0	3.4748(18)	3.475878(13)	−6.61920(5)	13.96951(15)
1	2.9093(16)	2.910571(13)	−5.10339(5)	9.95358(14)
2	2.2635(13)	2.264839(12)	−3.68159(5)	6.69457(12)
3	1.7200(11)	1.721335(12)	−2.60888(4)	4.45525(11)
4	1.3013(10)	1.302520(11)	−1.84985(4)	2.98472(10)

TABLE XIII. Coefficients ν_k of $\bar{u}^k$ in ν for $N = 0$–4, in units of 10^{-2}: ν_8 from the polynomials of Ref. [14] (their quoted errors added linearly) and from this work, and the new ν_9 and ν_{10} (from H_{10} and C_{10}). Generated from the ledger [check S2].

N	$10^2\nu_8$ [14]	$10^2\nu_8$	$10^2\nu_9$	$10^2\nu_{10}$
0	−3.89135(17)	−3.89149(4)	8.08899(8)	−18.1875(5)
1	−3.17852(17)	−3.17869(4)	6.13316(8)	−12.7571(5)
2	−2.41413(15)	−2.41430(4)	4.35887(7)	−8.4522(4)
3	−1.79044(14)	−1.79059(4)	3.04851(7)	−5.5437(4)
4	−1.32044(13)	−1.32058(4)	2.13758(7)	−3.6611(4)

TABLE XIV. Route B at $N = 1$. Top: the sums over the intermediate-scheme graphs at L loops of $\Gamma^{(4)}(0)$, which have $L+1$ vertices, and of $\Gamma^{(2)}(0)$ and $\partial_{p^2}\Gamma^{(2)}(0)$, which have L, in the normalization of Ref. [14], from our runs and from the per-graph values of Ref. [14] (their errors added in quadrature). Bottom: the ninth- and tenth-order coefficients from route A and from route B, which takes the sums of Ref. [14] below eight vertices and ours from eight to ten. z: the difference in units of the combined uncertainty. Generated from the stored runs and the ledger [check S4].

Sum	L	This work	Ref. [14]	z
$\Gamma^{(4)}$	7	13509.529(19)	13509.5443(3)	−0.79
$\Gamma^{(2)}$	8	6165.902(10)	6165.9013(5)	+0.12
$\partial_{p^2}\Gamma^{(2)}$	8	−543.6422(15)	−543.64217(15)	−0.01
$\Gamma^{(4)}$	8	84849.82(13)	—	—
$\Gamma^{(2)}$	9	32627.58(5)	—	—
$\partial_{p^2}\Gamma^{(2)}$	9	−3201.296(8)	—	—
$\Gamma^{(4)}$	9	559371.9(2)	—	—
$\Gamma^{(2)}$	10	185488.29(12)	—	—
$\partial_{p^2}\Gamma^{(2)}$	10	−19767.96(2)	—	—
Coefficient		Route A	Route B	z
β_9		−2.366420(6)	−2.3671(16)	+0.40
$10^3\eta_9$		−5.10339(5)	−5.121(15)	+1.16
$\eta_{\phi^2,9}$		−0.303695(3)	−0.3037(3)	+0.15
$10^2\nu_9$		6.13316(8)	6.134(8)	−0.08
β_{10}		4.78115(5)	4.781(8)	−0.01
$10^3\eta_{10}$		9.95358(14)	9.84(7)	+1.53
$\eta_{\phi^2,10}$		0.620920(18)	0.6206(15)	+0.22
$10^2\nu_{10}$		−12.7571(5)	−12.75(4)	−0.14

point graphs, the two-point graphs and their p^2 derivatives with eight to ten vertices, and took the sums with fewer vertices from the per-graph values of Ref. [14]. Table XIV compares the two routes. Our eight-vertex sums agree with those of Ref. [14], and the ninth- and tenth-order coefficients of the two routes agree within 1.5 standard deviations [check S4]. Because of the cancellations of route B (Sec. II A 1), its uncertainties are 90 to 500 times larger, 7×10^{-4} to 7×10^{-3} relative, and the comparison is limited by them. With the eight-vertex sums of Ref. [14] in place of ours, the differences become −1.2, +4.0, +1.4 and −1.2 standard deviations for β_9, η_9, $\eta_{\phi^2,9}$ and ν_9. Route B then carries the per-graph errors of Ref. [14] added in quadrature, which for η_8 give an uncertainty 3.3 times smaller than the one quoted there; measured with it, the η_8 of Ref. [14] differs from ours by +2.9 standard deviations, and the difference carries over to η_9.

D. Critical exponents

Resummed as described in Sec. II D, the series give the fixed point and the exponents of Table XV. The eighth-order coefficients of Ref. [14] and ours give exponents that differ by at most 1×10^{-6}, 1×10^{-3} of the resummation errors, with the same central parameters: the more precise eighth-order coefficient of η changes nothing at this level, and the table gives the two in one row where their digits coincide. From the eighth to the tenth order the resummation errors decrease, for $N = 1$ from 1.4141(48) to 1.4093(22) for $\bar{u}^*$, from 0.63024(94) to 0.62997(51) for ν, from 0.0326(26) to 0.0333(18) for η and from 0.791(17) to 0.803(14) for ω. Over $N = 0$–3 the errors decrease by factors of 1.4–1.8 for ν and 1.4 for η, and by factors that grow with N, 1.6–10.5 for $\bar{u}^*$ and 0.9–7.0 for ω (the error of ω does not decrease for $N = 0$). The coefficient of $\bar{u}^{11}$ in β moves the tenth-order $\bar{u}^*$ for $N = 1$ from 1.4106(21) (with β through $\bar{u}^{10}$) to 1.4093(22). Our estimates agree with those of Ref. [9] within the errors of the latter for every entry, and they share their pattern of deviations from the non-perturbative values: the ninth and tenth orders do not remove those deviations, but make them more visible.

In units of the combined standard deviation, the tenth-order ν differs from the non-perturbative values by +0.6, 0.0, −2.0 and −3.1 for $N = 0$–3: it agrees for $N = 0$ and 1 (for $N = 1$, $\nu = 0.62997(51)$ against 0.62997097(12) from the conformal bootstrap [41]) and is low for $N = 2$ and 3. The tenth-order η is low for every N (differences −1.5, −1.6, −1.8 and −2.0), and it increases with the order; a low η is a known feature of the fixed-dimension series [8, 9]. For η the optimum lies at the upper end of the range of α, but extending the range to $\alpha = 12$ moves η by at most 2×10^{-4}. For ω the differences, −2.7, −1.9, +0.1 and +10.4, change sign with N: the resummed ω depends on N less than the non-perturbative one. For $N = 3$ the successive truncations agree closely ($\omega = 0.78148(82)$) on a value well above the Monte Carlo result 0.759(2) [40], where the seven-loop resummation had already settled [9], and the same holds, to a lesser degree, for ν. The Monte Carlo values are not subject to the asymptotic nature of the series, and the conformal bootstrap, where it determines the same exponents, agrees with them to their quoted errors (Table XV). In these cases, then, our error estimates, which measure only the apparent convergence of the resummation, do not cover its systematic error.

a. Limitations and cross-checks. The deviations of Table XV point to a limitation specific to fixed dimension: the RG functions are expected to be non-analytic at $\bar{u}^*$, because of confluent corrections to scaling [1, 11, 81–

TABLE XV. Fixed point and critical exponents from the resummed series (Sec. II D 2). For each N, in the chronological order of the series: the fixed-dimension estimates of Ref. [9] (Table 1, $d = 3$; series through $\bar{u}^7$; for $N = 3$ the source prints $\omega = 0.782 \pm 0.0013$, an uncertainty ten times smaller than those of the other N, which we read as 0.782(13), as does Ref. [1]); the eighth-order series (β, η and ν through $\bar{u}^8$) with the coefficients of Ref. [14] and with ours, in one row where the two give the same digits (Sec. III D); our ninth-order series (through $\bar{u}^9$) and our tenth-order series (η and ν through $\bar{u}^{10}$, β through $\bar{u}^{11}$), all resummed with the same method; and non-perturbative values: self-avoiding walks for $N = 0$ ($\eta = 2 - \gamma/\nu$, $\omega = \Delta_1/\nu$), the conformal bootstrap for $N = 1$ ($\omega = \Delta_{\epsilon'} - 3$), and Monte Carlo simulations of improved lattice models and the conformal bootstrap for $N = 2$ and 3 (for the latter $\eta = 2\Delta_\phi - 1$, $\nu = 1/(3 - \Delta_s)$ and, for $N = 2$, $\omega = \Delta_{s'} - 3$). The errors of our results are those of the resummation, including that of $\bar{u}^*$; the statistical errors are at most 0.1 times as large (Sec. II D 3). Generated from the ledger [check S2].

N	Series	$\bar{u}^*$	ν	η	ω
0	7th order, Ref. [9]	1.413(6)	0.5882(11)	0.0284(25)	0.812(16)
	8th order, Ref. [14] and this work	1.4175(64)	0.58825(66)	0.0278(26)	0.798(24)
	9th order, this work	1.4137(54)	0.58806(57)	0.0281(20)	0.807(25)
	10th order, this work	1.4098(41)	0.58788(45)	0.0283(18)	0.818(26)
	self-avoiding walks, Refs. [77, 78]	—	0.58759700(40)	0.0310434(21)	0.899(14)
1	7th order, Ref. [9]	1.411(4)	0.6304(13)	0.0335(25)	0.799(11)
	8th order, Ref. [14] and this work	1.4141(48)	0.63024(94)	0.0326(26)	0.791(17)
	9th order, this work	1.4116(33)	0.63006(66)	0.0329(20)	0.796(15)
	10th order, this work	1.4093(22)	0.62997(51)	0.0333(18)	0.803(14)
	conformal bootstrap, Refs. [41, 79, 80]	—	0.62997097(12)	0.036297612(48)	0.82968(23)
2	7th order, Ref. [9]	1.403(3)	0.6703(15)	0.0354(25)	0.789(11)
	8th order, Ref. [14] and this work	1.4049(31)	0.6698(14)	0.0344(23)	0.785(11)
	9th order, this work	1.4038(12)	0.66998(94)	0.0347(17)	0.7869(54)
	10th order, this work	1.40297(50)	0.67019(74)	0.0352(16)	0.7895(31)
	Monte Carlo, Ref. [39]	—	0.67169(7)	0.03810(8)	0.789(4)
	conformal bootstrap, Ref. [42]	—	0.67175(10)	0.038176(44)	0.794(8)
3	7th order, Ref. [9]	1.390(4)	0.7073(35)	0.0355(25)	0.782(13)
	8th order, Ref. [14] and this work	1.3913(16)	0.7062(19)	0.0343(18)	0.7816(58)
	9th order, this work	1.39117(58)	0.7070(17)	0.0347(14)	0.7818(24)
	10th order, this work	1.39138(16)	0.7075(13)	0.0352(13)	0.78148(82)
	Monte Carlo, Ref. [40]	—	0.71164(10)	0.03784(5)	0.759(2)
	conformal bootstrap, Ref. [43]	—	0.71168(41)	0.03787(13)	—

83]. A non-analytic term of the form $(\bar{u}^* - \bar{u})^\Delta$ with non-integer Δ slows the convergence of any resummation that assumes analyticity there, and the successive truncations can agree with each other well before they agree with the exact value: apparent convergence, which is what the criteria of Sec. II D 2 measure, does not guarantee accuracy, and the derivative that gives ω is the most exposed. A longer series helps to quantify this systematic error but cannot remove it: from the eighth to the tenth order the uncertainties of ν, η and ω at $N = 1$ shrink only slowly (Table XV), and that of η, the exponent farthest from its non-perturbative value, no faster than that of ν, unlike in the ε expansion, where the uncertainty of η was found to fall rapidly with the loop order while those of ν and ω did not [34]. Other resummation methods can serve as cross-checks: Padé approximants of the Borel transform, as in the original analyses of the fixed-dimension series [2, 12, 13]; conformal mapping combined with a homographic transformation of the expansion parameter, with the parameters and the errors chosen by the criteria of recent resummations of the ε expansion [29, 34]; and variational perturbation theory [3], which in its strong-coupling form works with the bare coupling and determines the approach exponent ω self-consistently, and has been applied to the seven-loop series [84]. We have not applied them here: the first two share the assumption of analyticity at $\bar{u}^*$, and the comparison with the non-perturbative values already shows the size of the systematic error.

IV. SUMMARY AND OUTLOOK

We have extended the fixed-dimension RG functions β, η and ν of the three-dimensional $O(N)$ model to tenth order, and β to eleventh, for general N, and re-evaluated all lower orders, by sampling on GPUs the $\bar{R}$-subtracted parametric integral of every graph. At $N = 1$ the new coefficients have relative uncertainties between 2×10^{-6} and

4×10^{-5}; they agree with the large-N limit and with the large-order behaviour, and at the ninth and tenth orders with the independent route B, within its coarser precision. Through seven loops the per-order sums reproduce the published ones, and the eighth-order coefficients of η and ν are now 120 and 4 times more precise than before.

The design goal, a cost that grows slowly with the order, is met: from seven loops on, the time needed to integrate an order to a fixed relative precision grows by a factor of 2.4–2.9 per loop on average, whereas the number of graphs grows factorially. The bottleneck is now the construction of the integrands, which grows faster than the number of graphs and at the tenth order already costs more than the sampling. A sampler that draws graphs and points together, without enumerating the graphs, as the global tropical sampler of Ref. [54] does for unsubtracted integrands, would remove it once extended to the forest-subtracted integrands used here.

The resummed exponents are limited by the resummation, not by the coefficients: the statistical errors are at most 0.1 times the resummation errors, and the remaining deviations from the non-perturbative values, largest for ω at $N=3$, point to the non-analyticity of the RG functions at the fixed point. More orders, or resummations that account for this non-analyticity, are needed to go further.

Every per-class result is pooled in a public ledger, from which the generated numbers of this paper are rebuilt and checked (Appendix D). Further runs can be contributed and folded into it exactly, and a new order needs only its graph list and computing time.

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Appendix A: Conventions and normalizations

These conventions are also summarized, with the names used by the code, in the file `doc/CONVENTIONS.md` of the code repository (Appendix E).

a. Couplings. The renormalized coupling g is defined by $\Gamma^{(4)}_R(0)=m\,g$ [Eq. (2)], with the $O(N)$ tensor $(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})/3$ factored out. The integrals are expanded in $u=g/8\pi$ and the RG functions are quoted in

$$\bar u=\frac{N+8}{6}\,u=\frac{N+8}{48\pi}\,g,\qquad \beta(\bar u)\equiv m\frac{\partial\bar u}{\partial m}=\frac{N+8}{6}\,\beta_u, \tag{A1}$$

the normalized coupling of Refs. [12, 13], which is the $\tilde u$ of Ref. [14]. With it $\beta(\bar u)=-\bar u+\bar u^2+O(\bar u^3)$ for every N. We write η_{ϕ^2} for the anomalous dimension of ϕ^2 (η_2 in Ref. [15]), so that $1/\nu=2-\eta+\eta_{\phi^2}$, as in Ref. [14].

b. Integrals. Our integrals have one d^3k/π^2 per loop [17], that is 8π times the measure $d^3k/(2\pi)^3$; with this choice each loop comes with one power of u, the one-loop bubble at zero momentum equals one, and the low-order coefficients are rational. (In Ref. [15] "Nickel's normalization" refers to the coupling $\bar u$ rather than to the measure; integrals normalized with $d^3k/(2\pi)^3$ per loop are $(8\pi)^{-L}$ times ours.) The propagators of B_L, H_L and C_L have the renormalized mass, $m=1$, whereas those of Refs. [14, 15, 17], and of the intermediate-scheme integrals we use for comparison, have the mass of the intermediate scheme, $m_I=1$. Individual integrals in the two conventions differ by powers of m^2/m_I^2; the RG functions do not. Reference [14] quotes graph values V_G that include its weight factor and a factor $(16\pi)^L$ per graph with L loops, with the measure $d^3k/(2\pi)^3$. At $N=1$ the sum of its intermediate-scheme four-point values at L loops equals $(2^L/3)\sum_G S_G I_G$, where I_G is the integral of G in our normalization, in the same scheme and with $m_I=1$; this is the relation behind the comparisons of intermediate-scheme order sums in Sec. III A.

c. Weights. The weight $S_G=T_G(N)\,E!/|\mathrm{Aut}\,G|$ of Sec. II B includes the $O(N)$ factor $T_G(N)$, normalized to $T_G(1)=1$ as are the symmetry factors of Ref. [14]. The insertion of $\frac12\phi^2$ is normalized by $\Gamma^{(2,1)}_R(0;0)=1$, and $\partial\Gamma^{(a,b)}/\partial m_0^2=\Gamma^{(a,b+1)}$ at zero insertion momentum, so that, e.g., $C_1>0$.

d. From the per-order sums to the series. Writing $[\bar u^k]F$ for the coefficient of $\bar u^k$ in F, Eqs. (4), (6) and (A1) give, at each order,

$$\begin{aligned}
[\bar u^{L+1}]\beta &= 2(-1)^{L+1}\Big(\frac{6}{N+8}\Big)^L\big(B_L-2H_L\big)+\dots,\\
[\bar u^{L}]\eta &= 2(-1)^{L}\Big(\frac{6}{N+8}\Big)^L H_L+\dots,\\
[\bar u^{L}]\eta_{\phi^2} &= 2(-1)^{L}\Big(\frac{6}{N+8}\Big)^L C_L+\dots,\\
[\bar u^{L}]\nu &= \frac{(-1)^{L+1}}{2}\Big(\frac{6}{N+8}\Big)^L\big(C_L-H_L\big)+\dots,
\end{aligned} \tag{A2}$$

where the dots stand for products of sums of lower order ($H_1=0$). The coefficient of $\bar u^{L+1}$ in β thus depends on B_L and H_L, those of $\bar u^L$ in η on H_L, and in ν on C_L and H_L; the errors of the new sums enter linearly, with the prefactors of Eq. (A2). The reference values of B_L, H_L, C_L used in Sec. III are obtained by solving these

relations order by order for the sums, starting from the series of Ref. [14] [check S1].

At the lowest orders the sums are known exactly,

$$B_1 = \frac{N+8}{12}, \qquad B_2 = \frac{5N+22}{54}, \qquad H_2 = \frac{N+2}{243}, \\ C_1 = \frac{N+2}{12}, \qquad C_2 = \frac{N+2}{36}, \tag{A3}$$

that is 3/4, 1/2, 1/81, 1/4 and 1/12 at $N = 1$, and give

$$\begin{aligned} \beta(\bar{u}) &= -\bar{u} + \bar{u}^2 - \frac{4(41N+190)}{27(N+8)^2}\,\bar{u}^3 + O(\bar{u}^4), \\ \eta(\bar{u}) &= \frac{8(N+2)}{27(N+8)^2}\,\bar{u}^2 + O(\bar{u}^3), \\ \eta_{\phi^2}(\bar{u}) &= -\frac{N+2}{N+8}\,\bar{u} + \frac{2(N+2)}{(N+8)^2}\,\bar{u}^2 + O(\bar{u}^3), \\ \nu(\bar{u}) &= \frac{1}{2} + \frac{N+2}{4(N+8)}\,\bar{u} \\ &\quad + \frac{(N+2)(27N-38)}{216(N+8)^2}\,\bar{u}^2 + O(\bar{u}^3), \end{aligned} \tag{A4}$$

in agreement with Ref. [14]. At $N = 1$ the coefficients are $-308/729$ ($\bar{u}^3$ in β), $8/729$ ($\bar{u}^2$ in η), $-1/3$ and $2/27$ (η_{ϕ^2}), and $1/12$ and $-11/5832$ (ν). In the notation of Eq. (7), $p_{\beta,3} = -4(41N+190)/27$, $p_{\eta,2} = 8(N+2)/27$, $p_{\nu,1} = (N+2)/4$ and $p_{\nu,2} = (N+2)(27N-38)/216$.

Appendix B: Graphs

a. Counts. Table XVI gives, for every loop order, the numbers of 1PI graphs and of the integrands of Eq. (9) that do not vanish identically. One-vertex-reducible (1VR) graphs vanish for all three kinds, and so do, for H_L, the graphs with both external legs on one vertex (Sec. II A 3) [check I2]; through seven loops no other graph vanishes, so that the numbers of integrands are the numbers of the remaining graphs; at eight and nine loops the expansion confirmed that every remaining graph has a non-vanishing integrand (no class was dropped at the build). At ten loops the table gives the numbers of remaining graphs [check G5]. In the runs reported in Sec. III each class is a single graph.

b. Generator. The generation is described in Sec. II B 1. At eight loops the 26 540 four-point graphs have their legs distributed as $(2,2)$ (2254 graphs), $(2,1,1)$ (14 614) and $(1,1,1,1)$ (9672); at nine loops the 9184 two-point graphs as (2) (1804) and $(1,1)$ (7380) [check G1].

c. Twins. Besides the seven-loop numbers of Sec. II B 2, the four-point lists at five and six loops contain 104 and 518 distinct unsubtracted integrals for 129 and 660 graphs, and the eight-loop two-point list 1512 for 1622. Merging twins, after a pointwise check of their subtracted integrands, would save the corresponding samples.

d. Notation. A Nickel index [17] lists, for each vertex in turn, an `e` for every external leg and the labels of the vertices with equal or higher label to which it is joined, once per line, followed by a bar; the canonical index is the lexicographic minimum over the labellings of the vertices, with $| < \mathtt{e} < 0 < 1 < \dots$. For example, the one-loop four-point bubble is `ee11|ee|` and the two-loop sunset `e111|e|`. Our generator, like the one of Ref. [15] and the programs built on it [20], writes a graph instead as its number of vertices followed by the run-length-coded upper triangle of its symmetric matrix of line multiplicities, with optional sections for vertex colours (external legs) and self-loops. Those programs canonicalized it as a lexicographically minimal adjacency matrix, and their labels such as D40L5N78 (a four-point graph with five loops, line 78 of their list) are list positions, not graph invariants. For instance, the string `6(.2 1^2.2 1^6.1 1^2)` of D40L5N78 is the graph `ee12|345|345|e4|5|e|`. A table that maps every historical label to its Nickel index accompanies the graph lists (Appendix E).

Appendix C: Zero-dimensional sum rules

In zero dimensions every Feynman integral with unit propagators equals one. The sum $\sum_G S_G$ of the weights over a complete list is therefore the corresponding coefficient of the vertex functions of the zero-dimensional $O(N)$ model [15, 85, 86], with partition function

$$Z = \int d^N\phi\, \exp\Big[-\frac{m_0^2}{2}\,\phi^2 - \frac{g}{24}\,(\phi^2)^2\Big]. \tag{C1}$$

With $Q = \langle\phi^2\rangle/N$ and $A = \langle(\phi^2)^2\rangle/[N(N+2)]$ one has $\Gamma_0^{(2)} = 1/Q$ and $\Gamma_0^{(4)} = -3(A-Q^2)/Q^4$, and, as formal power series in g,

$$\langle(\phi^2)^k\rangle = \frac{I_k}{I_0}, \quad I_k = \sum_{n\ge0} \frac{(-g/24)^n}{n!}\,\frac{P_{k+2n}(N)}{m_0^{2(k+2n)}}, \tag{C2}$$

with $P_j(N) = N(N+2)\cdots(N+2j-2)$. For the lists without self-loops (one-vertex-reducible graphs included), the bare mass is shifted, $m_0^2 = 1 - (N+2)g/6$, so that every graph with a self-loop cancels against a graph with a mass counterterm; then

$$\begin{aligned} \Gamma_0^{(2)} &= 1 + \sum_{L\ge2} (-1)^{L+1}\Sigma_{2,L}\, g^L, \\ \Gamma_0^{(4)} &= \sum_{L\ge0} (-1)^{L}\Sigma_{4,L}\, g^{L+1}, \end{aligned} \tag{C3}$$

with $\Sigma_{E,L} = \sum_{G\in\mathcal{G}_E(L)} S_G$. With $m_0^2 = 1$ the same expansion gives the sums over all 1PI graphs, self-loops included. Table XVII gives the polynomials $\Sigma_{E,L}(N)$. Through eight loops they are those of the zero-dimensional computation of Ref. [15], whose table stops at $\Gamma^{(4)}$ with seven loops and $\Gamma^{(2)}$ with seven, and whose original output extends to eight loops in both. The

TABLE XVI. Numbers of 1PI graphs per loop order L and of non-vanishing integrands. $|\mathcal{G}_4(L)|$, $|\mathcal{G}_2(L)|$: graphs without self-loops, one-vertex-reducible ones included; 1VR: one-vertex-reducible graphs among them; one vertex: two-point graphs with both legs on one vertex (generalized tadpoles), 1VR ones included; B, H, C: integrands of Eq. (9) that survive the algebraic pruning, i.e. graphs that are not 1VR and, for H, do not have both legs on one vertex; with self-loops: complete two-point lists including graphs with self-loops, used only for checks. [check G1]

L	$\lvert\mathcal{G}_4(L)\rvert$	1VR	B	$\lvert\mathcal{G}_2(L)\rvert$	1VR	one vertex	H	C	with self-loops
1	1	0	1	0	–	–	–	1[a]	1
2	2	1	1	1	0	0	1	1	2
3	8	2	6	2	0	1	1	2	5
4	27	8	19	6	1	2	4	5	17
5	129	33	96	19	3	7	11	16	68
6	660	158	502	75	13	21	50	62	313
7	3986	836	3150	317	52	85	209	265	1632
8	26 540	4851	21 689	1622	235	356	1157	1387	9685
9	197 022	30 975	166 047[b]	9184	1146	1804	6776	8038	63 774
10	1 593 254	215 576	1 377 678[b]	59 614	6217	10 064	46 102[b]	53 397[b]	463 933

[a] The one-loop tadpole with two insertions.
[b] Graphs that are not 1VR (and, for H, do not have both legs on one vertex); all their integrands survive the pruning, and every one of them was sampled [check G5].

printed table has two sign misprints, absent from the original output: the last four terms of the six-loop $\Gamma^{(2)}$ numerator and the term in N^3 of the three-loop $\Gamma^{(2,1)}$ numerator have the same sign as the other terms. The nine- and ten-loop rows were obtained here from Eqs. (C1)–(C3), which reproduce every coefficient of the original computation at lower orders.

The lists used here reproduce every row of Table XVII exactly, as polynomials in N, through ten loops for both $\Gamma^{(4)}$ and $\Gamma^{(2)}$, up to the 1 593 254 ten-loop four-point graphs; the $O(N)$ factor of every graph is counted exactly by carrying the partial pairings of its vertices from vertex to vertex, which agrees graph by graph with the enumeration of all 3^V pairings wherever both run [check G2]. The lists with self-loops reproduce the bare series, whose coefficients at $N = 1$ are 1/2, 5/12, 5/6, 115/48, 625/72, 10 775/288, 80 275/432, 21 615 925/20 736, 67 531 025/10 368 and 620 085 875/13 824 at $L = 1$–10. They also reproduce the sum rules of $\Gamma^{(2,1)}$ of Ref. [15], obtained by inserting ϕ^2 on any line of a graph without self-loops and on the self-loop of a graph with exactly one, at five and six loops for all N, the seven-loop polynomial of the original output, and its eight-loop value at $N = 1$.

Appendix D: Reproducibility

Every claim of this paper that a script can verify is marked in the text with the identifier of its check, as in [check G2]. The checks are part of the code repository (Appendix E), in its directory `checks/`: one script per identifier, which states the claim as it is made here, its inputs in the repository and the expected result, prints the numbers it obtains, and ends with `PASS` or `FAIL`. Table XVIII lists them. The command

```
make -C checks
```

runs them all and writes a report, `checks/REPORT.md`; `make -C checks quick` runs those that take less than about a minute, and `python3 checks/run.py G2` runs one. They need Python 3.6 or later with its standard library, GNU make and a C++17 compiler with OpenMP, which build the CPU integrator and the compiled helpers on first use. Where a check needs what the machine lacks, a GPU for the live part of N2 or the MPMATH library for I1, it reports `SKIP` for that part and verifies instead the stored outputs of the runs that did it, which the repository keeps (`results/`).

The numbers that describe the pooled Monte Carlo data are not typed in the sources of the paper. The bodies of Tables III, V, VII–XIII, XIV, XV and X, the rows of this work in Table VI, Figs. 1 and 2, and the values quoted from them in the text are written by `make -C series` from the ledger of pooled per-class statistics (Sec. II C 4) and from the stored runs of route B. Check S2 regenerates all of them except Table XIV, which check S4 regenerates, and both verify that they are identical to those printed here. The exceptions are the numbers of individual early runs and tests in Sec. III A (the round-off example of H_4 and the test of fused multiply-add contraction), which are historical; where a check exists, it verifies them from their stored outputs.

Appendix E: Data and code availability

The code, the data and the checks are in the repository ReBar, https://github.com/PaoloRibeca/ReBar, whose release 0.4.3 is archived at Zenodo, https:

TABLE XVII. Zero-dimensional sum rules: $\Sigma_{E,L}(N) = \sum_{G\in\mathcal{G}_E(L)} S_G$ over the L-loop 1PI graphs without self-loops, one-vertex-reducible graphs included [Eq. (C3)], written as a numerator polynomial over a denominator D, and its value at $N = 1$. Rows to eight loops: Ref. [15] (printed table and original output); the nine- and ten-loop rows: this work, from the lists and from the exact zero-dimensional series, which agree. [check G2]

E	L	Numerator	D	$N=1$
4	1	$8+N$	6	3/2
	2	$108+26N+N^2$	36	15/4
	3	$2048+724N+62N^2+N^3$	216	105/8
	4	$47648+21632N+2928N^2+124N^3+N^4$	1296	893/16
	5	$1297664+702864N+127392N^2+9044N^3+220N^4+N^5$	7776	8795/32
	6	$40362304+24950944N+5549104N^2+546408N^3+23404N^4+358N^5+N^6$	46656	97987/64
	7	$1410726400+966932160N+250483648N^2+31004768N^3+1898240N^4+53312N^5+546N^6+N^7$	279936	10951025/1152
	8	$54747705344+40784744192N+11897634304N^2+1746112576N^3+137234272N^4+5654416N^5+110208N^6+792N^7+N^8$	1679616	49985915/768
	9	$2336870225920+1865178040064N+598669644160N^2+100305560704N^3+9494392864N^4+511500848N^5+14961264N^6+211032N^7+1104N^8+N^9$	10077696	2245562203/4608
	10	$108856221154304+92103242480128N+31990963159296N^2+5962312144640N^3+652186499392N^4+42906530592N^5+1667820720N^6+36012864N^7+379788N^8+1490N^9+N^{10}$	60466176	4057808535/1024
2	2	$2+N$	18	1/6
	3	$10+7N+N^2$	54	1/3
	4	$280+232N+52N^2+3N^3$	648	7/8
	5	$2552+2384N+688N^2+71N^3+2N^4$	1944	211/72
	6	$113888+116608N+39952N^2+5680N^3+320N^4+5N^5$	23328	3413/288
	7	$1498912+1650832N+641936N^2+112904N^3+9260N^4+316N^5+3N^6$	69984	48323/864
	8	$90828928+106190272N+45557472N^2+9353504N^3+982000N^4+51060N^5+1134N^6+7N^7$	839808	3123017/10368
	9	$778012480+956252384N+444020448N^2+102678120N^3+12884816N^4+882730N^5+31028N^6+473N^7+2N^8$	1259712	9443467/5184
	10	$118963074560+152597681664N+75647848448N^2+19237783296N^3+2767156512N^4+231239968N^5+10977808N^6+273456N^7+2982N^8+9N^9$	30233088	506798407/41472

//doi.org/10.5281/zenodo.23750242; this paper is arXiv:2610.39307. The code under the GNU General Public License, version 3 or later, the data and the documentation under the Creative Commons Attribution 4.0 International licence, and the text of this paper under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International licence. The repository is the Supplemental Material of this paper; its file `SUPPLEMENTAL.md` says where each of the following items is.

1. *Graph lists*: $\Gamma^{(4)}$ and $\Gamma^{(2)}$ to ten loops, in the notation of Refs. [15, 20] (`graphs/lists/`) and, in the form the integrator reads (`integrator/data/lists/`), with the canonical Nickel index, the weight $E!/|\mathrm{Aut}\,G|$ at $N = 1$ and the one-vertex-reducible and self-loop flags of every graph; the $O(N)$ weight polynomials are in the ledger, class by class, with the SHA-256 of every list the integrator ran on. The lists to seven ($\Gamma^{(4)}$) and eight ($\Gamma^{(2)}$) loops coincide with those of Ref. [14], version 1.0.2 (check G3), and the four six- and seven-loop lists of the integrator are its files, with its columns, reproduced under its MIT licence; the $\Gamma^{(4)}$ lists from eight loops and the $\Gamma^{(2)}$ lists from nine loops on are new.

2. *Per-class results* (`ledger/`): the statistics of every class pooled over the runs (number of samples, mean and central moments to the fourth, extremes, round-off indicators, time per sample), the exact weight polynomials, the per-order sums with the reference values and the pulls, and a record of every run with its own statistics (Sec. II C 4); and, in the archive deposited with the release (too large for the repository), the output of every run as received, with the integral, error, samples and diagnostics of every class. With them the coefficients can be recombined for any N, the error propagation of Sec. II D 3 can be repeated, and two runs can be compared class by class.

3. *Run records*: dated snapshots of the series and tables derived from the ledger (`results/coefficients/`), and the reference data and inputs of the checks (`results/checks/`). The complete records of the campaigns of Sec. III (reports, logs and environment records) are kept outside the repository; the run files of the ledger's contributions are deposited with its archive.

TABLE XVIII. The reproducibility checks of the repository (directory `checks/`). Where: the sections and appendices whose claims cite the check. Needs: CPU, a computation on a CPU with the graph tools or the CPU build of the integrator; GPU, a computation on a GPU; stored, only the stored outputs of the repository (the ledger and the reference data of the checks).

Id	Where	Claim	Needs
G1	II B 1, II B 3, App. B	Every graph list holds distinct, connected 1PI graphs in canonical Nickel form, in the numbers of Tables II and XVI	CPU
G2	II B 3, II E, App. C	The lists satisfy the zero-dimensional sum rules exactly, as polynomials in N, through ten loops, and reproduce every row of Table XVII	CPU
G3	II B 3	The lists equal those of Ref. [14] as sets: $\Gamma^{(4)}$ to seven loops, $\Gamma^{(2)}$ to eight	CPU
G4	II B 3	The weights and $O(N)$ polynomials equal those of Refs. [15, 20] on all graphs with $L \leq 5$ and on the four-loop tables	CPU
G5	II C 2, App. B	The counts of Table XVI follow from the lists (1VR by a cut-vertex test, checked against the lists' flags), and the ledger holds that many classes for every order with data	CPU
I1	II B 3, II E	The expanded forest formula, Eq. (13), agrees pointwise with an independent evaluation in 100-digit arithmetic	CPU[a]
I2	II B 2, II B 3, II E, App. B	One-vertex-reducible graphs, graphs with self-loops and generalized tadpoles vanish after the pruning	CPU
I3	II B 2, II B 3, II E	Twins (2-isomorphic graphs) have pointwise equal subtracted integrands	CPU
I4	II B 2	A run that loads the built classes from the cache is bitwise identical to one that builds them	CPU
N1	II C 5	Our Philox4x32-10 reproduces the known-answer vectors of Ref. [74]	CPU
N2	II C 6, II E, III A	On identical random numbers the GPUs reproduce the per-class means of the CPU to round-off (golden runs)	CPU, GPU[b]
N3	II B 3, III A	The exact two-loop sums $B_2 = 1/2$, $H_2 = 1/81$ and $C_2 = 1/12$ are reproduced within the errors	stored
N4	III A	The sums at three to seven loops agree within the errors with those derived from Ref. [14]; the statistics of the pulls	stored
N5	II C 4, III A	The ledger is rebuilt bit for bit from the archive of its runs, and every run is consistent class by class with the pool of the runs before it (per-class χ^2)	stored[c]
N6	III A	FP64 round-off does not bias the order sums: paired double and extended-precision evaluation of 55 classes (the 43 most exposed of B_8, H_8, C_8, H_9, C_9 and C_{10}, and 12 controls) finds no significant difference, and at most 4×10^{-6} of an order's error summed over its classes	CPU
N7	II C 2, III A	Eight replicas of each of B_6, H_6, C_6, B_7, H_7 and C_7 scatter as their quoted errors say, order sums and class by class	stored
N8	III A	The round-off scales of the four classes of H_4 quoted as an example in Sec. III A	stored
S1	II E, III, App. A	The Callan–Symanzik relations map the sums derived from Ref. [14] back to its coefficients, to 10^{-15}	stored
S2	II C 4, II D 2, II D 3, II E, III A, III B, III C, III D	Every number of the generated tables and values is regenerated from the ledger by `make -C series`	stored
S3	II B 3, II E	The large-order ratios of Table X follow from the coefficients and the constants of Sec. II E	stored
S4	II E, III C	Route B reproduces the coefficients of Ref. [14] from its per-graph values, its eight-vertex sums agree with those of Ref. [14] and its ninth- and tenth-order coefficients with route A (Table XIV)	stored
S5	II E, III C	The terms in N^L of every order sum vanish class by class, as the limit $N \to \infty$ requires, and those in N^{L-1} of every order of the ledger agree with the $1/N$ expansion, which reproduces the leading coefficients of Ref. [14]; the polynomials $p_{X,k}(N)$ have the degrees and leading coefficients that follow	stored

[a] With the MPMATH library; without it, the check verifies the stored values (`results/checks/`).
[b] Without a GPU, the check verifies the stored GPU outputs (`results/checks/`).
[c] The replay needs the archive of the runs (`ledger-archive/`, outside the repository); without it, the check verifies the invariants of the ledger and reports the replay as skipped.

4. *Series* (`series/`): the tools that validate and pool the runs and derive from the ledger the polynomials $B_L(N)$, $H_L(N)$, $C_L(N)$ and $p_{X,k}(N)$, the coefficients and the generated tables of this paper; the polynomials $p_{X,k}(N)$ with the full covariance of their coefficients (Sec. III C), in `series/generated/polynomials.json` and, one row per coefficient, `polynomials.tsv`, with a dated copy in `results/coefficients/`.

5. *Checks* (`checks/`): the scripts of Appendix D.

6. *Code* (`integrator/`, `graphs/`): the program that builds and integrates the subtracted integrands (C++17 and CUDA from a single source; versions 0.3.0 to 0.4.3 for the runs of Sec. III, each recorded with its build signature in the ledger), with its self-tests and validation driver, and, in `results/checks/`, the golden reference files of Sec. II C 6; the graph generator of Sec. II B 1; and the scripts that convert the graphs to Nickel indices and evaluate the sum rules of Appendix C.

Every results archive records the hardware, driver and compiler versions and the checksums of all sources. Since the random numbers of a class depend only on the seed, the kind of integrand, the loop order and the Nickel index (Sec. II C 5), any class of any run can be recomputed independently: bit for bit with the same build on CPUs, for any number of threads, and within round-off with other compilers and libraries and on GPUs (Sec. II C 6).

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